

2020



#بالحبّ نغرس.. فكرة.. جذورها.. أنتم

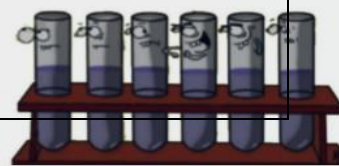
General Biology 102

Done by: Hala Khanjer

References :

Campbell biology-11th edition

& Dr-Qotaiba's Records



EVIDENCE FROM STUDYING METABOLIC ACTIVITY

قبل ما ندخل ونحكي كيف الجينات بتعمل البروتينات السؤال هو ؟ كيف اكتشفوا العلاقة بينهم

IN 1902 , British physician **Archibald Garrod** was the first to suggest that genes dictate phenotypes through enzymes (proteins that catalyze chemical reaction in the cell). He postulated that the symptoms of an inherited disease reflect an ability to particular enzyme . He referred to such diseases as “inborn errors of metabolism “ as **alkaptonuria** . Garrod reasoned that people with this disease have an inherited ability to make an enzyme which are going to break down alkapton .

Inborn errors of metabolism: disease form a large class of **genetic diseases** that cause disorders of **metabolism**. The majority are due to defects of single **genes** that code for **enzymes** that facilitate conversion of various substances (**substrates**) into others (**products**).

Alkaptonuria : rare genetic disease that let people who are affected by it have black urine because they are unable to make an enzyme which will break alkapton , and alkapton will be expelled to urine.

زمان كان في كثير من الامراض التي لا يعرف سببها والعلماء يحاولوا أن يجدوا لها تفسير معين لربما استطاعوا علاجها ومن بعض الامراض الوراثية التي جعلتنا نتعرف على الجينات والبروتينات وأحد هاي الامراض هي : **inborn error of metabolism**

امراض وراثية تؤدي الى خلل في العمليات الايضية بسبب خلل في جين لم ينتج انزيم يعمل على تحويل مواد الى مواد أخرى ومن هذه الامراض **Alkaptonuria**

وهو مرض وراثي ناتج عن عدم قدرة انتاج جين لانزيم معين لتكسير مادة **Alkapton** مما يجعلها تتأكسد وتعطي لونا داكنا عند خروجها مع البول .

Several decades later, research supported Garrod's hypothesis that a gene dictates the production of a specific enzyme, later named the one gene-one enzyme hypothesis. Biochemists learned that cells synthesize and degrade most organic molecules via metabolic pathways, in which each chemical reaction in a sequence is catalyzed by a specific enzyme .

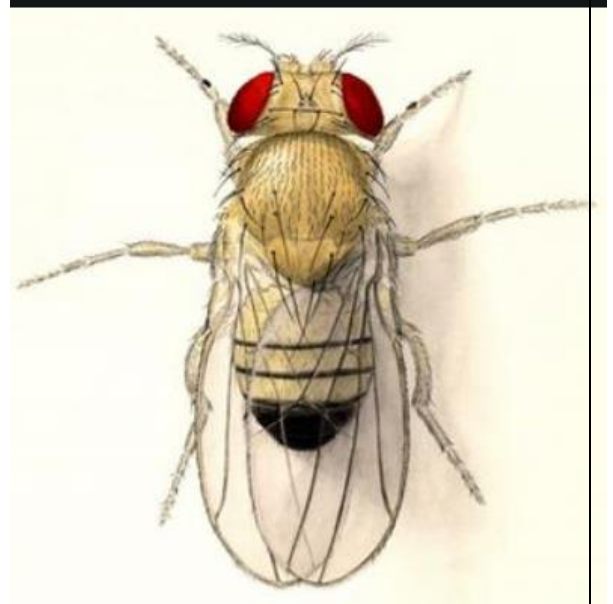
Such metabolic pathways lead, for instance, to the synthesis of the pigments that give the brown donkey its fur .color or fruit flies (*Drosophila*) their eye color (In the 1930s, the American biochemist and geneticist George Beadle and his French colleague Boris Ephrussi speculated that in *Drosophila*, each mutation affecting eye color blocks pigment synthesis at a specific step by preventing production of the enzyme that catalyzes that step. But neither the chemical reactions nor the enzymes that catalyze them were known at the time.

بعد عدة عقود و هنالك الكثير من الأبحاث التي دعمت كلام العالم (جين واحد يعمل على انتاج انزيم)
وسميت الفرضية one gene –one enzyme وعلماء البيوكيم علموا بأن الخلية (المصنع) بتكسر
المواد العضوية عن طريق Metabolic pathway بحيث كل علمية كيميائية تتسرع وتتحفز من خلال
انزيم معين . بعض الأنزيمات تعمل على صناعة صبغة في الحمير مما يعطيها اللون البني في فرو
الحمير . كما ان العالمان Beadle and Ephrussi افترضوا أن هناك طفرة حدثت أثناء تصنيع أحد
الانزيمات التي تصنع صبغة معين مما أدى إلى عدم وجود لون في أعين دروسوفيل

NUTRITIONAL MUTANTS IN NEUROSPORA . SCIENTIFIC INQUIRY



NOT



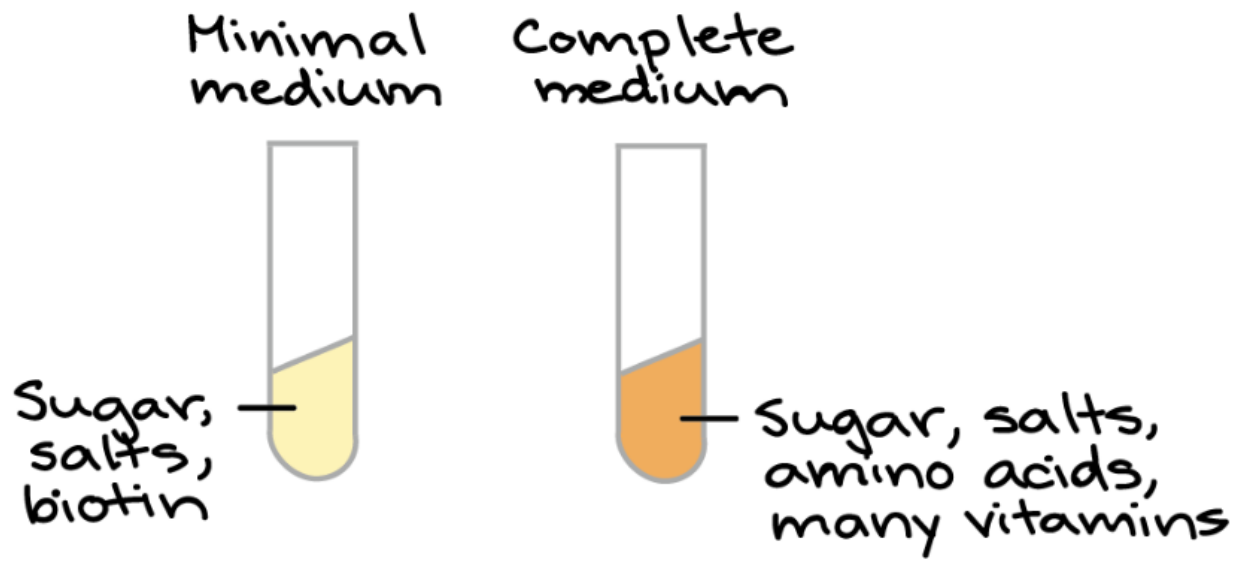
Beadle and Tatum worked with a simple organism: common bread mold, or *Neurospora crassa*. Using *Neurospora*, they were able to show a clear connection between genes and metabolic enzymes. Why bread mold is great for experiments

But why did Beadle and Tatum choose to study at mold not drosophila .

Well...at first, Beadle had planned to work with the fruit fly *Drosophila* (also a bit gross, but a much more common organism for experiments at the time). However, as he got more and more interested in the connection between genes and metabolism, he realized that *Neurospora* might give him a better way to answer the questions. For one thing, *Neurospora* had a fast and convenient life cycle, one with both haploid and diploid phases that made it easy to do genetic experiments.

بالأول ليه العلماء قرروا أنهم يعملوا على mold (العفن) مش على دروسوفيليا يلي كانت سائدة عليها التجارب بهاد الوقت ؟ الجواب بسيط لانهم لاحظوا بأنه هي دورة حياتها سريعة ما بتحتاج لمراحل نمو كثيرة مقارنة بالدروسوفيليا وهي تعتبر

Haploid : one set of chromosomes and diploid : two sets of chromosomes



Minimal and complete medium Neurospora wild type could grow In minimal it will be able to synthesis any amino acid that will let to survive (because amino acids are the building block to make the proteins which are important for all species to survive and forming visible colonied f genetically identical cells . Also , in complete medium it will be happy a lot because it will survive and save its energy .

العفن اللطيف قادر أنه ينمو بميديا (بتحتوي على امينو اسيد) وما بتحتوي ؟ كيف هيك ؟! الموضوع بسيط إذا كانت ما بتحتوي على امينو اسيد بتصير في عمليات حيوية و عدة خطوات لانتاج الامينو اسيد يلي هي بحاجة ويلي هو يساعد على البقاء ، وإذا لقت الامينو أسيد يلي بدها هاااي بتكون وفرت استهلاك طاقتها وأخذته واستخدمته لحتى يساعدها تضل عايشة .

طيب من هاي المعلومات كيف العالمين بدهم يتأكدوا إذا في ربط بين الجينات والانزيمات ؟ عن طريق عمل طفرات إذا هالطفرة ما صنعت انزيم إذا هي مش حتقدر تنمو بالتالي حتموت . بس كيف حيعملوا الطفرة ؟

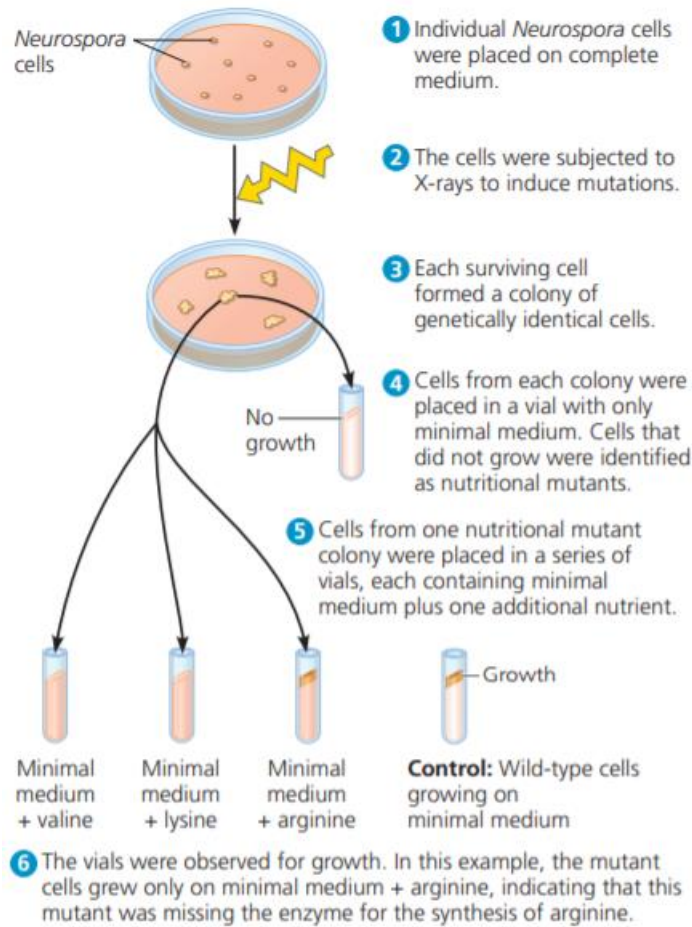
By bombarding the Neurospora with X-rays known to cause mutation , and they looked among the survivors for mutants that differed in their nutritional needs from the wild type bread mold , and looked among the survivors for mutants that differed in their nutritional needs .

عملوا على تعريضها لاشعة اكس راي المعروف انها تسبب طفرات بعد ذلك نظروا للاختلاف بينها وبين ال Wild type التي لم تعرض لأكس راي ولم يحدث لها طفرات كانت النتيجة بانه التي عرضت لطفرات لن تستطيع النمو في

Minimal media

Wild type : refers to the phenotype of the typical form of a species as it occurs in nature.

يعني هي زي ما تواجدت بالطبيعة بدون أي طفرة .



- 1) تم وضع خلايا تيروسبورا في ميديا بيحتوي على جميع العناصر الغذائية الملائمة إضافة للأمينو أسيد complete medium
- 2) تم تعريض الخلايا لاشعة اكس راي مما سبب طفرات والخلية التي لم تمت بسبب الطفرات تكاثرت ونمت و عملت خلايا متطابقة جينيا مثلها Colony
- 3) كل Colony أخذنا منها مجموعة خلايا ووضعناها Minimal media (ما يحتوي على امينو اسيد) الخلايا لم تنمو والسبب معناته انها ما قدرت تصنع المواد الغذائية بسبب الطفرات يلي حدثت . لا تنسوا كيف عرفنا انه السبب الغذاء لانه قبل كان موجود بميديا فيو كل الامينو اسيد والاشياء يلي بتخليه ياخذ غذائه بدون ما يتعب أما هلا نقلنا لميديا هو بصنع غذاءه فيها , وكما ان مرة لا تنسوا زمان كان ينمو ب هلا اذا ما نمت معناته ما تصنع البروتين المناسب وكيف عرفنا انه البروتين علاقته بالجين لانه بهداك الوقت كان معروف انه اشعة اكس راي بتاثر عالجينات . فاذا النيروسبورا بس وديناها عميديا مافي غذاء وما قدرت تتكاثر معناته هي بكل بساطة مش قادرة تصنع انزيمات يلي حتصنع الغذاء .
- 4) بعد هيك وضعنا كل خلية Minimal medium مع واحد من الامينو أسيد ولاحظنا كمان أنه مش بكل الامينو أسيد نمت فهون ما نمت الا لنا ضعفنا ارجنين و هاد بيعني انه نيرسبورا صار عندها خلل بتصنيع انزيم يلي يساعد على انتاج ارجنين وبس ضعفنا الارجنين نمت وتكاثرت .

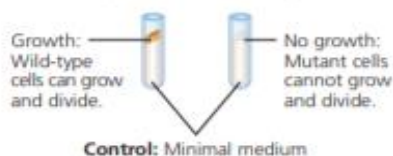
السؤال المهم ؟ هل arginine يتصنع بخطوة واحدة ولا عدة خطوات بتحتاج انه

مادة تحول لمادة لحتى نوصل لل arginine ؟ هون أدى لتجربة

ثانية (لا تخافوا من الرسمة والجدول بسيطين هلا حنشرحهم) .

Experiment Working with the mold *Neurospora crassa*, Adrian Srb and Norman Horowitz, then at Stanford University, used Beadle and Tatum's experimental approach (see Figure 17.2) to isolate mutants that required arginine in their growth medium. The researchers showed that these mutants fell into three classes, each defective in a different gene. From studies by others on mammalian liver cells, they suspected that the metabolic pathway of arginine biosynthesis involved a precursor nutrient and the intermediate molecules ornithine and citrulline, as shown in the diagram on the right.

Their most famous experiment, shown here, tested both the one gene-one enzyme hypothesis and their postulated arginine-synthesizing pathway. In this experiment, they grew their three classes of mutants under the four different conditions shown in the Results Table below. They included minimal medium (MM) as a control, knowing that wild-type cells could grow on MM but mutant cells could not. (See test tubes below.)



Results As shown in the table on the right, the wild-type strain was capable of growth under all experimental conditions, requiring only the minimal medium. The three classes of mutants each had a specific set of growth requirements. For example, class II mutants could not grow when ornithine alone was added but could grow when either citrulline or arginine was added.

Results Table				
Condition	Classes of <i>Neurospora crassa</i>			
	Wild type	Class I mutants	Class II mutants	Class III mutants
Summary of results	Can grow with or without any supplements	Can grow on ornithine, citrulline, or arginine	Can grow only on citrulline or arginine	Require arginine to grow

Conclusion From the growth requirements of the mutants, Srb and Horowitz deduced that each class of mutant was unable to carry out one step in the pathway for synthesizing arginine, presumably because it lacked the necessary enzyme, as shown in the table on the right. Because each of their mutants was mutated in a single gene, they concluded that each mutated gene must normally dictate the production of one enzyme. Their results supported the one gene-one enzyme hypothesis, proposed by Beadle and Tatum, and also confirmed that the arginine pathway described in the mammalian liver also operates in *Neurospora*. (Notice in the Results Table that a mutant can grow only if supplied with a compound made after the defective step because this bypasses the defect.)

Data from A. M. Srb and N. H. Horowitz, The ornithine cycle in *Neurospora* and its genetic control, *Journal of Biological Chemistry* 154:129-139 (1944).

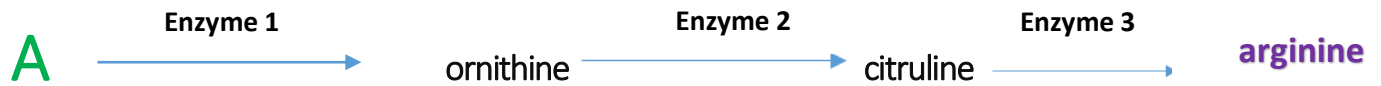
Gene (codes for enzyme)	Wild type	Class I mutants (mutation in gene A)	Class II mutants (mutation in gene B)	Class III mutants (mutation in gene C)
Gene A	Enzyme A	Enzyme A	Enzyme A	Enzyme A
Gene B	Enzyme B	Enzyme B	Enzyme B	Enzyme B
Gene C	Enzyme C	Enzyme C	Enzyme C	Enzyme C
	Precursor → Ornithine → Citrulline → Arginine	Precursor → Ornithine → Citrulline → Arginine	Precursor → Ornithine → Citrulline → Arginine	Precursor → Ornithine → Citrulline → Arginine

WHAT IF? > Suppose the experiment had shown that class I mutants could grow only in MM supplemented by ornithine or arginine and that class II mutants could grow in MM supplemented by citrulline, ornithine, or arginine. What conclusions would the researchers have drawn from those results regarding the biochemical pathway and the defect in class I and class II mutants?

هلاً بدنا نفهم مفهومين ؟

PRECURSOR : is a compound that participates in a chemical reaction that produces another compound.

INTERMEDIATE : intermediate is a substance formed during a middle step of a chemical reaction between reactants and the desired product.



Precursor

Intermediate

Product

الجدول هاد بشرحنا كيف **mold** ما بتتكاثر بسبب عدم قدرتها على انتاج Arginine فحسب ال

Wild type زي ما عرفنا بنموا بكل مرحلة لانه لو في نقص فالجينات حتنصع انزيمات وتاخذ حاجتها لكن لو

نفترض بأنه هي ما بتصنع الارجنين مثل . Class I كانت لما حطيناها Minimal meduim ما نمت لكن mutant

لما ضفنا Ornithin نمت معناته العائق يلي ما بخليها تصنع Arg هو تصنع orth إذا في خلل بتصنيع

هاد الأنزيم 1 طيب اذا في خلل لنفكر شو سببه مش احنا عرضناها لاشعة اكس راي والاشعة هاي كنا منعرف

انها بتأثر على الجينات اذا صار خلل في جين 1 يلي بصنع انزيم 1 يلي من خلاله حنعمل orth

لو نفس المثال اعطيناه ctr كمان ممتاز لانه حيتحول لارجنين.

وهاي هي كل الفكرة كل مرة بدنا نعطيها مادة ونشوف اذا نمت او لا بس تنمو بالمادة يلي اعطيناها ياها معناته في خلل بتصنيع الانزيم يلي يحول المادة الأولى للثانية بسبب خلل في جين , ولا تنسوا إذا اعطيناها ارجنين نكون ريحنا العفن المسكين واخذ يلي بده ياه ونما .

DON'T FORGET : beadle and tatum set up their conditions so that each mutant was defective in a single gene.

Example for this hypothesis is albino donkey because it lacks a key enzyme called tyrosine to produce melanin (dark pigment) . The absence of melanin causes white fur as well as its ears , hooves , eyes are pink because no melanine is present to mask the reddish color of blood vessels that run through this structure .

مثال على هذه الفرضية هو نوع من أنواع الحمير (البينو) سبب بياض الفرو والاذنين والعينين مائلين للزهري بسبب عدم وجود صبغة الميلانين



The Products of Gene Expression: A Developing Story

They made revisions to the one gene-one enzyme hypothesis : First of all , not all proteins are enzyme . **Keratin** , the structural protein of animal hair , and hormone **insulin** are two example nonenzyme proteins .

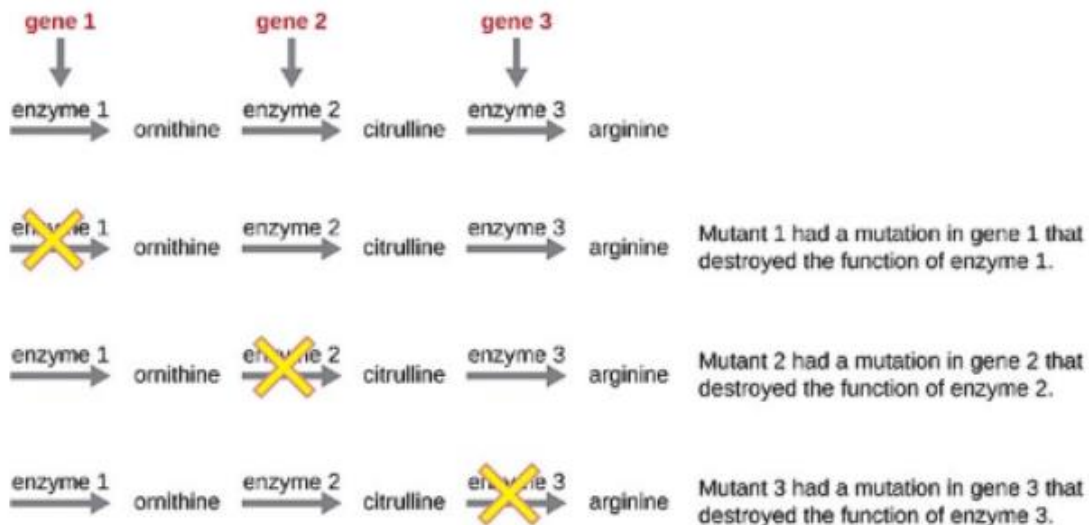
Second , many proteins are made from 2 or more different polypeptides is specified by its own gene , like hemoglobin (2 kinds of polypeptides) .

Three , alternative splicing and you will learn it in this chapter .

أولا انه ليس كل البروتينات هي انزيمات وليس كل انزيمات هي بروتينات ومثال على ذلك الكيراتين .
 ثانيا بعض البروتينات مصنوعة من أكثر من سلسلة امينو أسيد مثل الهيموجلوبين .
 ثالثا ونحنكي عنه بالتفصيل ان شاء الله .

هاد الجدول بيحي زي ه بالامتحان وبحكيك تكتب التسلسل فيه مع اجابته

Beadle and Tatum Experiments				
Bread Mold	Minimal Medium (MM)	MM + Ornithine	MM + Citrulline	MM + Arginine
Wild type	grew	grew	grew	grew
Mutant 1	did not grow	grew	grew	grew
Mutant 2	did not grow	did not grow	grew	grew
Mutant 3	did not grow	did not grow	did not grow	grew



REVISION

The study of metabolic defects provided evidence that genes specify proteins.

In 1909, Archibald Gerrod was the first to suggest that genes dictate phenotype through enzymes that catalyze specific chemical reactions in the cell.

He suggested that the symptoms of an inherited disease reflect a person's inability to synthesize a particular enzyme.

He referred to such diseases as "inborn errors of metabolism."

Gerrod speculated that alkaptonuria, a hereditary disease, was caused by the absence of an enzyme that breaks down a specific substrate, alkapton.

Research conducted several decades later supported Gerrod's hypothesis.

Progress in linking genes and enzymes rested on the growing understanding that cells synthesize and degrade most organic molecules in a series of steps, a metabolic pathway.

In the 1930s, George Beadle and Boris Ephrussi speculated that each mutation affecting eye color in *Drosophila* blocks pigment synthesis at a specific step by preventing production of the enzyme that catalyzes that step.

However, neither the chemical reactions nor the enzymes that catalyze them were known at the time.

Beadle and Edward Tatum were finally able to establish the link between genes and enzymes in their exploration of the metabolism of a bread mold, *Neurospora crassa*.

They bombarded *Neurospora* with X-rays and screened the survivors for mutants that differed in their nutritional needs.

Wild-type *Neurospora* can grow on a minimal medium of agar, inorganic salts, glucose, and the vitamin biotin.

Beadle and Tatum identified mutants that could not survive on minimal medium, because they were unable to synthesize certain essential molecules from the minimal ingredients.

However, most of these nutritional mutants can survive on a complete growth medium that includes all 20 amino acids and a few other nutrients.

One type of mutant required only the addition of arginine to the minimal growth medium.

Beadle and Tatum concluded that this mutant was defective somewhere in the biochemical pathway that normally synthesizes arginine.

They identified three classes of arginine-deficient mutants, each apparently lacking a key enzyme at a different step in the synthesis of arginine.

They demonstrated this by growing these mutant strains in media that provided different intermediate molecules.

Their results provided strong evidence for the one gene—one enzyme hypothesis.

Later research refined the one gene—one enzyme hypothesis.

First, not all proteins are enzymes.

Keratin, the structural protein of hair, and insulin, a hormone, both are proteins and gene products.

This tweaked the hypothesis to one gene—one protein.

Later research demonstrated that many proteins are composed of several polypeptides, each of which has its own gene.

Therefore, Beadle and Tatum's idea has been restated as the one gene—one polypeptide hypothesis.

Some genes code for RNA molecules that play important roles in cells although they are never translated into protein.

BASIC PRINCIPLE OF TRANSCRIPTION AND TRANSLATION

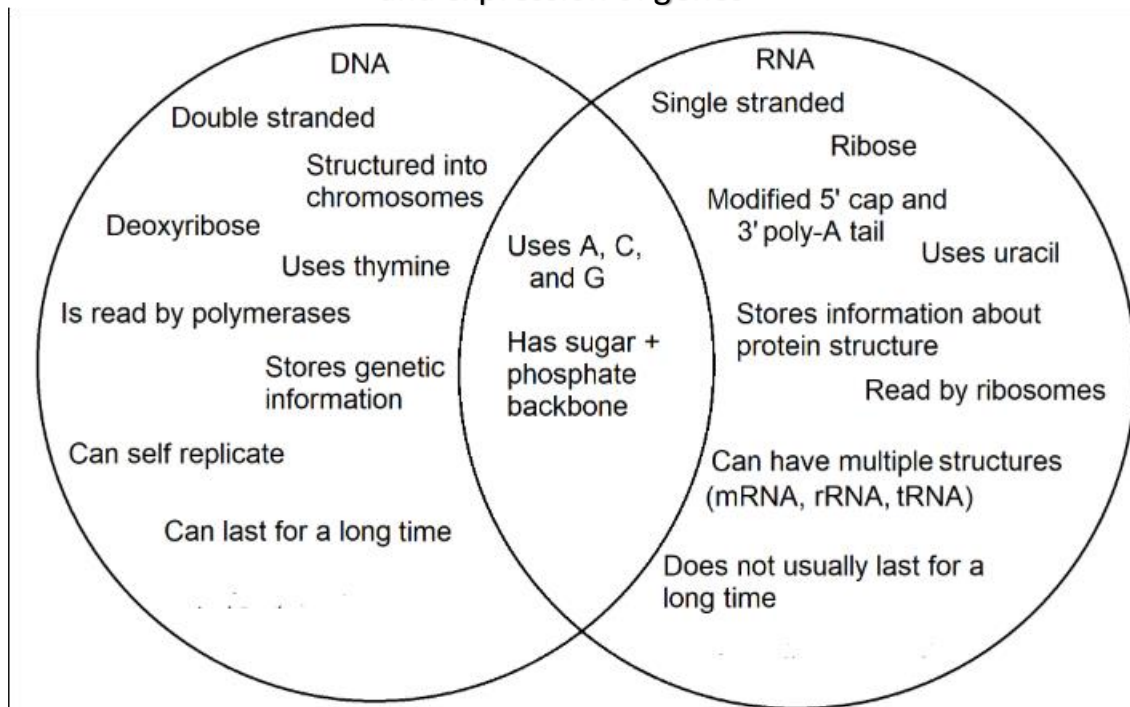
هاد الكونسبت ما حلتزم بالكتاب حشرحه وأقيم أي حشو وإذا في أشياء ما لقيتوه فحكون حاكيتيه بباقي الأجزاء مثل الفرق بخطوات الترجمة والنسخ بين البكتيريا وحقيقيات النوى تمام .

Gene : the basic physical and functional unit of heredity. Genes are made up of DNA. Some genes act as instructions to make molecules called proteins. However, many genes do not code for proteins.

الجينات هي الوحدة الفيزيائية والوظيفية للوراثة وهي جزء من الحمض النووي , بعض أنواع الجينات تعطي أوامر لصنع بروتينات والبعض الآخر لا , فيكون له وظائف أخرى مثل تنظيم عملية صنع البروتينات .

DNA : s a molecule composed of two polynucleotide chains that coil around each other to form a double helix carrying genetic instructions for the development, functioning, growth and reproduction of all known organisms and many viruses.

RNA : is a polymeric molecule essential in various biological roles in coding, decoding, regulation and expression of genes



TYPES OF RNA :

حيمر علينا كله بالشابتر فبتقدروا تتركوا وترجعوله تمام

واخر 2 بشابتر 18 معنا .

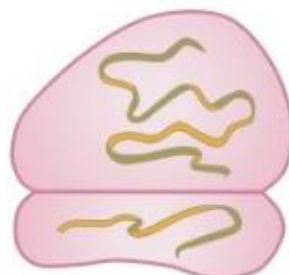
- RNA molecules are classified according to their structure and function.

Table 25.3 The Roles of Different Kinds of RNA

RNA Type	Size	Function
Transfer RNA	Small	Transports amino acids to site of protein synthesis
Ribosomal RNA	Several kinds—variable in size	Combines with proteins to form ribosomes, the site of protein synthesis
Messenger RNA	Variable	Directs amino acid sequence of proteins
Small nuclear RNA	Small	Processes initial mRNA to its mature form in eukaryotes
Micro RNA	Small	Affects gene expression; important in growth and development
Small interfering RNA	Small	Affects gene expression; used by scientists to knock out a gene being studied



Messenger RNA (mRNA)



Ribosomal RNA (rRNA)



Transfer RNA (tRNA)

THE GENETIC CODE TABLE

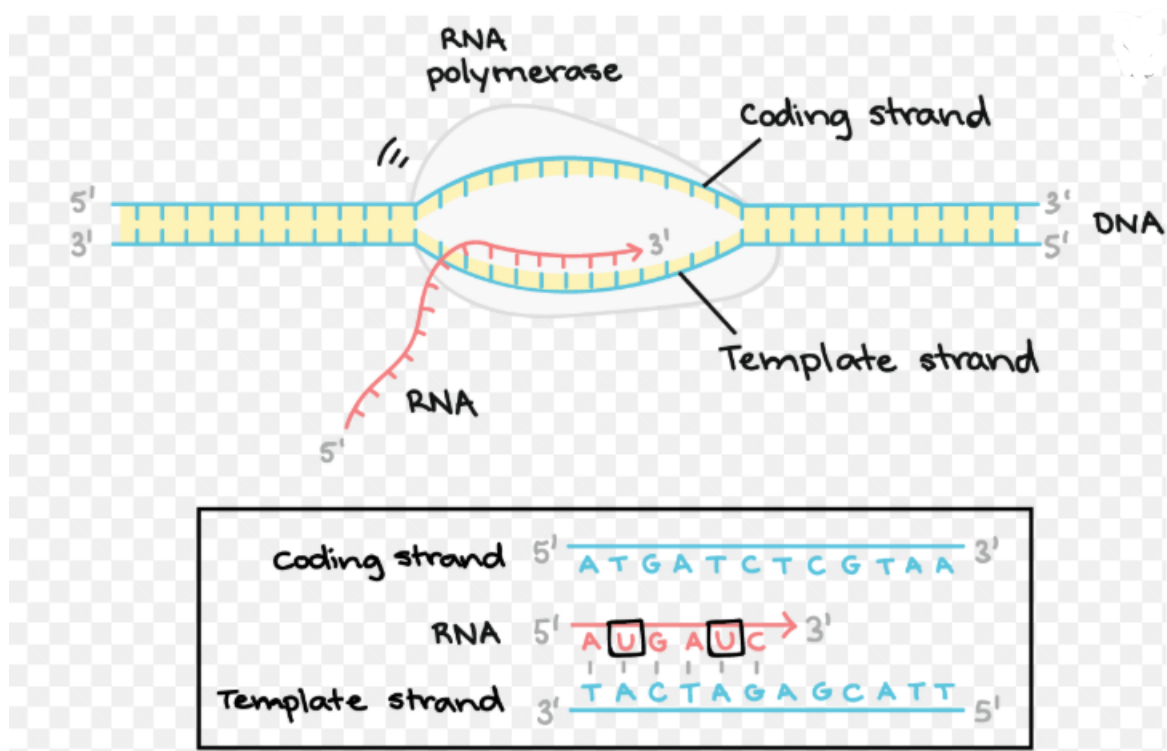
		Second letter				
		U	C	A	G	
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

خلينا ناخذ كم ملاحظة : هاد الجدول بيربط بين النيكلوتيدات والامينو اسيد
وبدنا نلاحظ انه الكودون بعبر عنه ب3 نيكلوتيدات وفي عنا أكثر من كودون
بيعطي لنفس الامينو اسيد

REMEMBER

Template strand or anti sense : is the term that refers to the strand used by DNA polymerase or RNA polymerase to attach complementary bases during DNA replication or RNA transcription.

Coding strand or sense : is the DNA strand whose base sequence corresponds to the base sequence of the RNA transcript produced (although with thymine replaced by uracil)

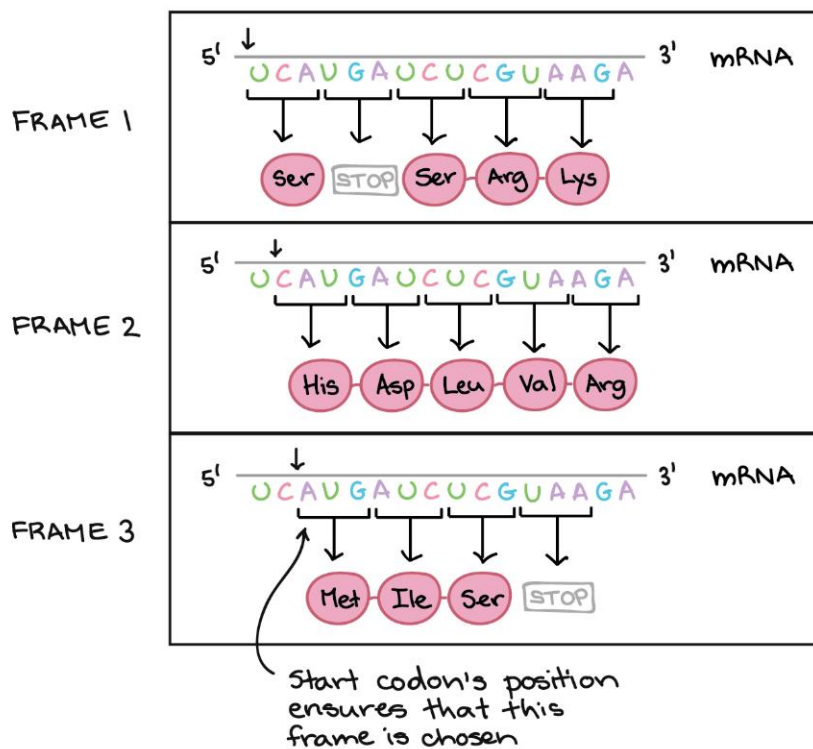


In transcription, the DNA sequence of a gene is copied to make an RNA molecule. This step is called transcription because it involves rewriting, or transcribing, the DNA sequence in a similar RNA "alphabet." In eukaryotes, the RNA molecule must undergo processing to become a mature messenger RNA (mRNA).

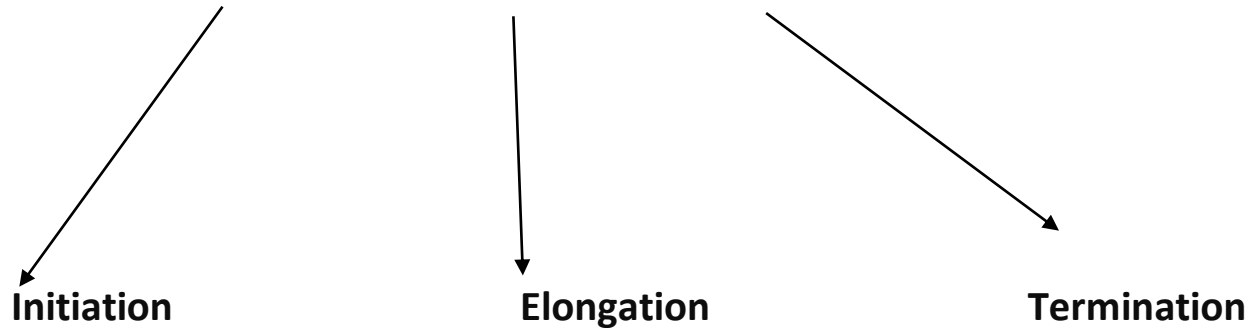
In translation, the sequence of the mRNA is decoded to specify the amino acid sequence of a polypeptide. The name translation reflects that the nucleotide sequence of the mRNA sequence must be translated into the completely different "language" of amino acids.

A gene that encodes a polypeptide is expressed in two steps. In this process, information flows from DNA $\rightarrow$ RNA $\rightarrow$ protein, a directional relationship known as the central dogma of molecular biology.

Reading Frame : is a way of dividing the sequence of nucleotides in a nucleic acid (DNA or RNA) molecule into a set of consecutive, non-overlapping triplets



SYNTHESIS OF AN RNA TRANSCRIPT



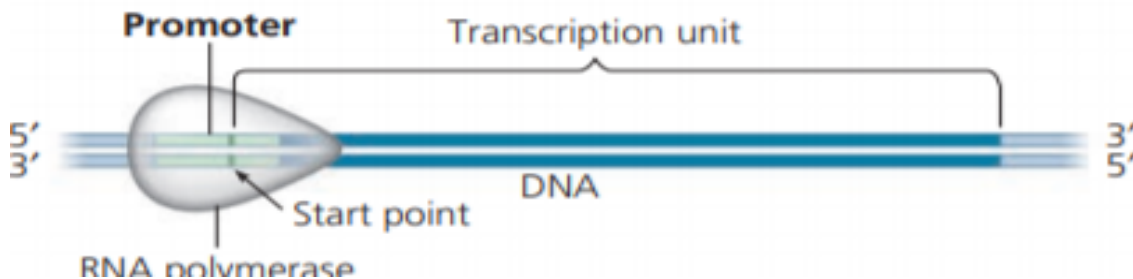
Initiation : is the beginning of transcription. It occurs when the enzyme RNA polymerase binds to a region of a gene called the promoter.

هي تعتبر أول خطوة في عملية ترانسكريبشن تحدث لما يرتبط **Rna polymerase** بالمنطقة محددة بالجين وهي **.promoter**

What is the difference in binding the RNA polymerase in bacteria and eukaryotes ?

In bacteria , part of RNA polymerase itself recognize and binds to the promoter .

بالبكتيريا في منطقة **RNA polymerase** ذات نفسها بتعرف وين لازم ترتبط بال **Promoter**



In eukaryotes , transcription factors are going to recognize the TATA box , must bind to the DNA before RNA polymerase II can bind in the correct position and orientation , the enzyme unwind the two DNA strands and begins transcribing at the **start point** .

هون الوضع بختلف عن البكتيريا حيخوا عنا مجموعة بروتينات تساعد هاد الانزيم يرتبط
بطريقة صحيحة يلي همي Transcription factor حيرتبطوا ب TATA BOX

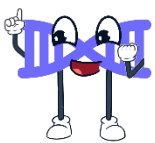
وبعدھا حيخي RNA polymerase يرتبط ويبدأ فك 2 ستراند ويبدأ الترجمة عند

Start point

TRANSCRIPTION INITIATION COMPLEX : complex from transcription factors and RNA polymerase II bound in the promoter .

The interaction between rna polymerase and transcription factors is an example **TO PROTEIN -PROTEIN INTERACTION**

TATA BOX : crucial promoter dna sequence which form the initiation complex at eukaryotic promoter .



WHAT IF ?

Suppose X-ray caused a sequence change in the TATA box of particular gene's promoter ?
How would that affect transcription of a gene ?

The transcription factor that recognize the TATA sequence would be unable to bind , so RNA polymerase could not bind and transcription that gene probably would not occur .

Elongation : is the stage when the RNA strand gets longer, due to the addition of new nucleotides .

RNA polymerase moves along the DNA , it untwists the double helix , exposing 10-20 nucleotides at a time for pairing with RNA nucleotides , in the advancing wave of RNA synthesis , the new RNA molecule peels away from its DNA template , and the DNA double helix re-form.

Transcription progresses at a rate about 40 nucleotides per second in eukaryotes .

أول اشي RNA Polymerase ارتبط مزبوط Promoter هالأ بدنا RNA strand أطول كيف ؟ عن طريق
. نضيف نيكليوتيدات . طول ما RNA polymerase بتحرك حيفك DNA (حوالي 10 ل 20)
. كل ما صار أطول حيطلع من DNA بالتالي DNA حيرجع يكون رابطة هيدروجينية ويرجع يرتبط مرة ثانية
. عملية إضافة النيكليوتيدات بالثانية عند حقيقيات النواة هي 40 بالثانية

انتبهوا : NOTES :

RNA polymerase walks 3-----5 strand but the transcription is from 5-----3 , RNA transcript is similar to coding strand instead that thymine will replace to uracil .

For a single gene it could transcribes by many RNA polymerases so it will increase the amount of RNA transcript so the cell will produce protein in large amounts .

Termination : The process of ending transcription is called termination, and it happens once the polymerase transcribes a sequence of DNA known as a terminator.

العملية التي تنتهي فيها عملية النسخ وتحدث لما يتم نسخ تسلسل من Terminator تعرف

In **bacteria** , transcription proceeds through **terminator sequence** in the DNA . The transcribed terminator (RNA sequence) function as termination signal causing the polymerase to detach from the DNA and release transcript which requires **no further modification** .

بالبكتيريا , حتى تتوقف عملية النسخ يكون في Terminator sequence على DNA .
هاد الاشي بترجم على RNA وظيفته زي الإشارة يلي تخلي Polymerase ينفصل من وبالتالي يطلع عنا جزيء RNA بدون أي تعديل .

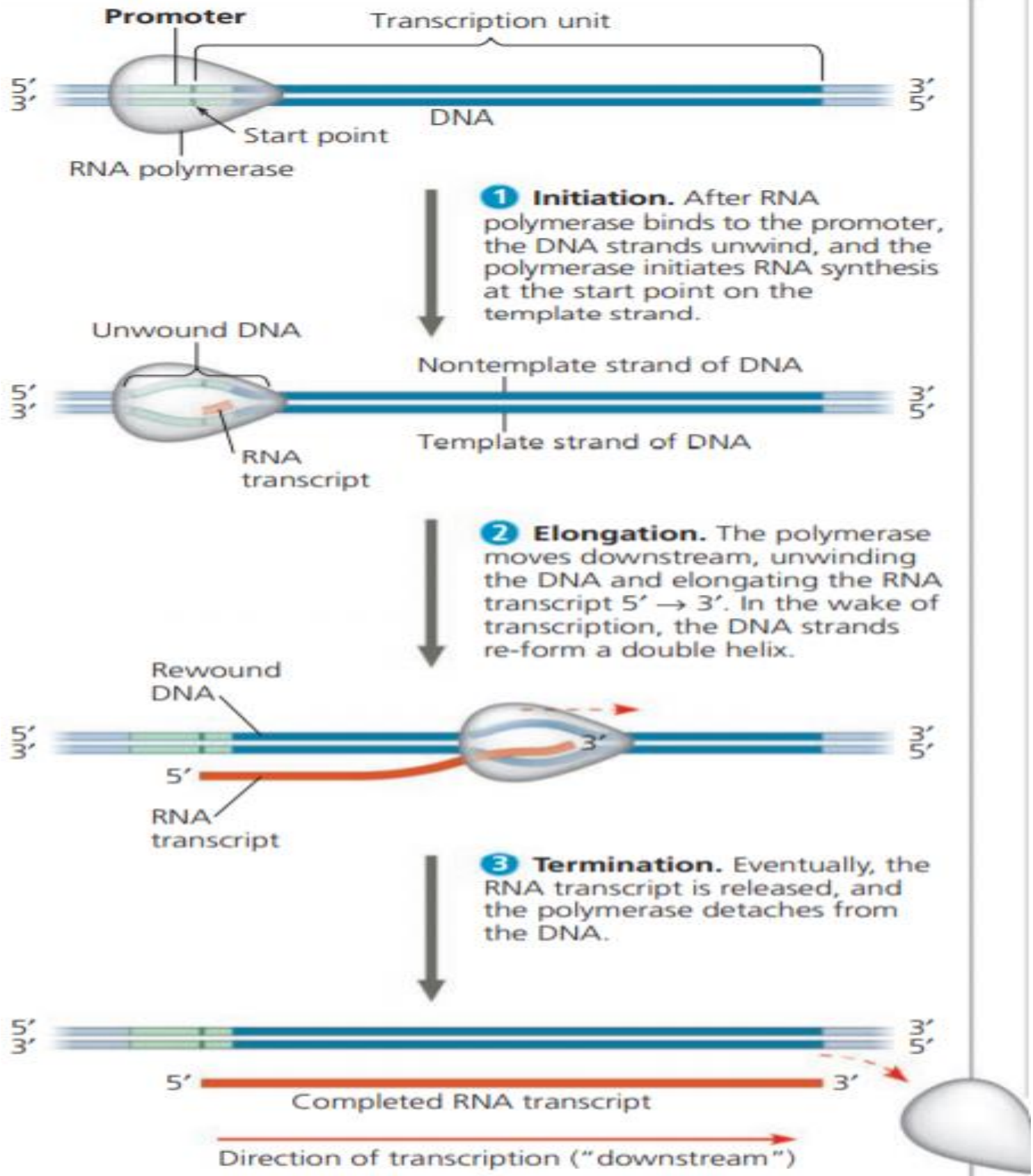


ركزوووووووووا كم مرة حكيت بدون تعديل ياااا ويلكم تخربطوا

In **eukaryotes** , RNA Polymerase II transcribes a sequence on DNA called **polyadenylation signal sequence in m RNA** , This is called a signal because once this stretch of six RNA nucleotides appears , it is immediately bound by certain proteins in nucleus . Then at a point about 10-35 nucleotides downstream from the polymerase , the proteins will cut the RNA transcript free from the polymerase , releasing the pre-m RNA , which **will have modification** .

في حقيقيات النواة , في سيكونس عنا بترجم من DNA هاي تعتبر زي إشارة لأنه بمجرد ما تظهر هاي النيوكليوتيدات حترتبط ببروتينات داخل النواة , تقريبا بعد 10-35 نيكلوتيد من Polymerase البروتينات حينقطع من DNA يعطينا Transcript **حيحتاج تعديل** .

TRANSCRIPTION باختصار



Eukaryotic cells modify RNA after transcription .

1) Alteration of m RNA Ends

Each end of pre-m RNA molecule is modified in a particular way . The 5'end , **which is synthesized first**, receives a 5'end after transcription of the first 20-40 nucleotides . The 3'end of pre- m RNA molecule is also modified before the pre- m RNA exist the nucleus ,Recall that the pre-m RNA is cut and released soon after the **polyadenylation signal ,AAUAAA**, is transcribed . At the 3'end , an enzyme then adds 50-250 more adenine (A) **forming poly -A tail** .

في حقيقيات النواة وبما أنه عندهم نواة إذا وهمي داخل النواة قبل ما يطلع Pre- m RNA حيصير تعديل على نهاياته , فمن جهة 5 حينضاف كمية جوانين (تقريبا اول 20-40) .

أما نهاية 3 كمان حيصير عليها تعديل قبل ما تطلع من النواة حينضاف حوالي 20-250 ليعطينا

• Poly a tail

What is the importance 5'cap and 3'poly A tail ?

First , they help the m RNA to export from the nucleus .

Second , they protect the m RNA from degradation by **hydrolytic enzyme** .

Third , they help the ribosome attach to 5'end of the m RNA once the m RNA reaches the cytoplasm .

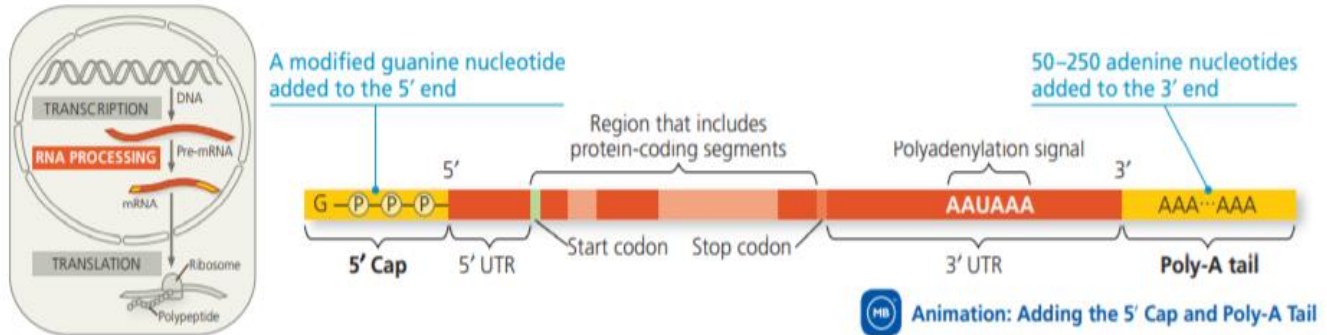
ليه مهمين اذا ؟ بتساعده ليطلع من النواة للسيتوبلازم , بتحميه من التكسر .
كمان بتساعد الريبوسومات اول ما يطلع انه يرتبط بنهاية 5 أول ما يوصل للسيتوبلازم .

▼ **Figure 17.11 RNA processing: addition of the 5' cap and poly-A tail.**

Enzymes modify the two ends of a eukaryotic pre-mRNA molecule. The modified ends may promote the export of mRNA from the

nucleus, and they help protect the mRNA from degradation. When the mRNA reaches the cytoplasm, the modified ends, in conjunction with certain cytoplasmic proteins, facilitate ribosome attachment. The 5' cap and poly-A

tail are not translated into protein, nor are the regions called the 5' untranslated region (5' UTR) and 3' untranslated region (3' UTR). The pink segments are introns, which will be described shortly (see Figure 17.12).



شوووو ضل ما حكيما عنه ؟؟؟ ركزوا بالرسمه شوي ايوااا

UTR : untranslated region refers to either of two sections, one on each side of a coding sequence on a strand of mRNA. If it is found on the 5' side, it is called the 5' UTR (or leader sequence), or if it is found on the 3' side, it is called the 3' UTR (or trailer sequence), it has many functions but the main function is ribosome binding .

هي مناطق لا تترجم لبروتينات موجودة عكلا 5 و 3 والها كثير من الوظائف منها ارتباطها بالريبوسوم .

Split genes and RNA splicing :

Introns : are non-coding regions of an RNA transcript, or the DNA encoding it, that are removed by splicing before translation , and it is located between coding region (exons) .

Exons is any part of a gene that will encode a part of the final mature RNA produced by that gene after introns have been removed by RNA splicing, (not all exons are translated regions because UTR it is not translated but inside the exons.

Pre – m RNA ال يحتوي على منطقتين وحدة بترجم للبروتينات exon بغض النظر انها بتحتوي على UTR
والثانية لا تترجم للبروتينات لكن الها فوائد ثانية منحكي عنها بعدين بس هاي Introns منشيلها عن طريق عملية RNA –
Splicing

RNA Splicing : is a form of RNA processing in which a newly made precursor messenger RNA (pre-mRNA) transcript is transformed into a mature messenger RNA (mRNA). During splicing, introns (Non-coding regions) are removed and exons (Coding Regions) are joined together.

How is the pre-m RNA splicing carried out ? **By spliceosome** (large complex made of proteins and small RNA , this complex released the introns and bind the 2 exons

What is the benefit of small RNA ? Firstly , participate in spliceosome assembly ,splice site recognition but also **catalyze splicing reaction** .

يعني هون منقدر نحكي مفصل كيف الانترون بنفصل ويربط 2 اكسون مع بعض

عن طريق Spliceosome مركب بيحتوي على بروتينات و SRNA هو يلي بفصل الاكسون من الانترون

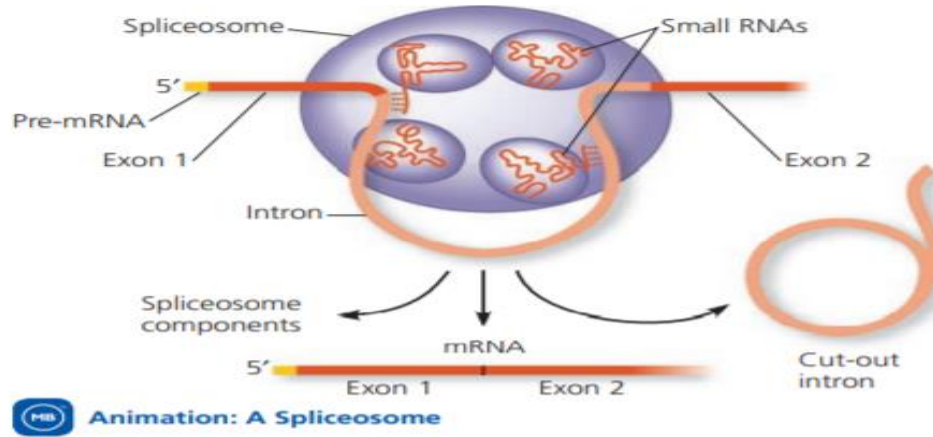
طبيب شو وظيفة هاد SRNA ؟ يساعد في تكوين Spliceosome وبتعرف عنمنطقة يلي لازم ينقص فيهم انترون

(مش غريب هالاشي) !

Splice recognition وأهم اشي بتسرع

فقرة

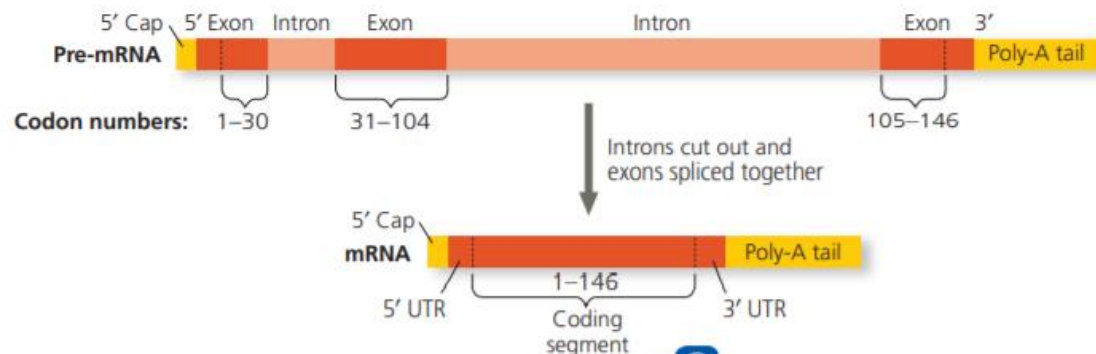
▼ **Figure 17.13 A spliceosome splicing a pre-mRNA.** The diagram shows a portion of a pre-mRNA transcript, with an intron (pink) flanked by two exons (red). Small RNAs within the spliceosome base-pair with nucleotides at specific sites along the intron. Next, small spliceosome RNAs catalyze cutting of the pre-mRNA and the splicing together of the exons, releasing the intron for rapid degradation.



ركزولي بالرسمه يلي تحت في اكثر من طريقة الاكسون ينشال ممكن 1 و 3 يكونوا مع بعض وممكن 1 و 2 و 3 معناته بصير في عنا تنوع وهاد الاشئ بسموه

Alternative RNA processing : s a regulated process during gene expression that results in a single gene coding for multiple proteins.

وطبعا اختلاف ترتيب الاكسونات مع بعض شو يؤدي انه بصير في عنا تنوع بالبروتينات لهيببيك البروتينات اكثر من الجينات



Molecular components of translation

شو رأيكم نتخيل اشي ؟ انا وصلتني مسج بس ياالله المسج لغته غير عني شو بدي أعمل ؟! أكب المسج ؟
أكيد لا!!!!!! بروح لعمو جوجل يترجمها وهون مين عنا المسج ومين المترجم .

Message : series of codon along an m RNA molecule , **Translator** is called a transfer RNA (t RNA).

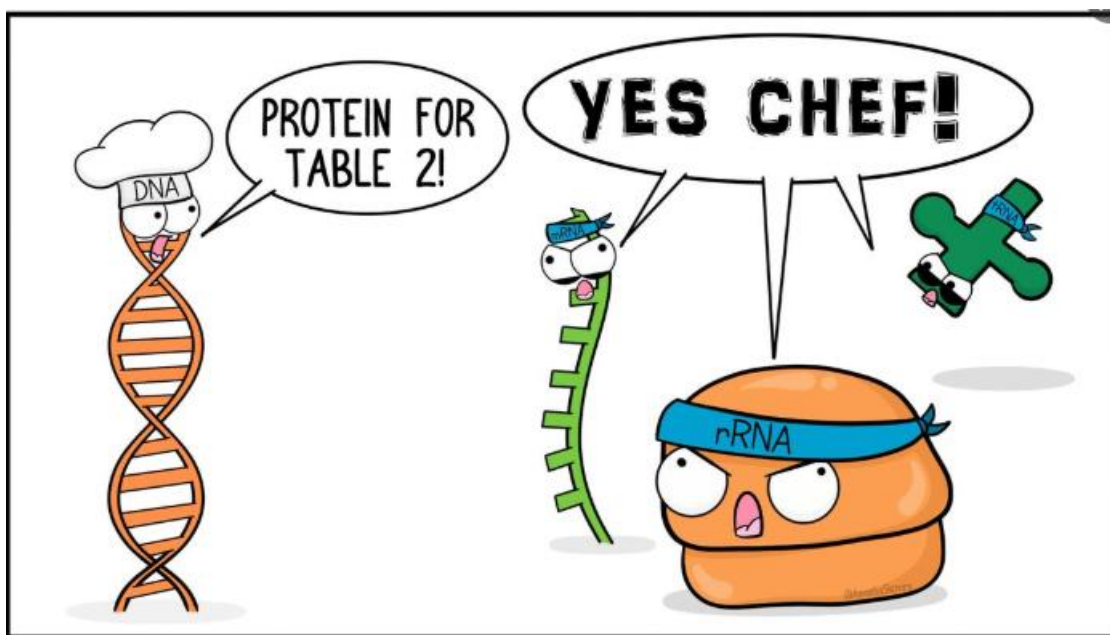
هلا خلينا ندا ب **T RNA**

Function of t RNA : transfer an amino acid from the cytoplasm pool of amino acids to growing polypeptide in ribosome .

A cell keeps its cytoplasm stocked with all 20 amino acids , From where ?

Either by synthesizing them from other compounds or by taking them up from surrounding solution .

Ribosome is made from proteins and RNA adds each amino acid brought to it by tRNA to the growing end of polypeptide chain.



THE STRUCTURE AND FUNCTION OF TRANSFER RNA

The t RNA molecules looks like a cloverleaf , it consists of a single RNA strand that is only about 80 nucleotides long (compared of hundreds of nucleotides in m RNA) . Because the presence of complementary stretches of nucleotides bases that can make hydrogen bonding to each other and forming by twisting and folding forming 3-dimensional shape , with 3' and 5' ends. (L shape)

The protruding 3' end acts as an attachment site for an amino acids ,the loop extending from other end of the L includes the anticodon .

جزء t RNA يتكون من 80 نيكليوتيد مقارنة بمئات النيكليوتيدات ب m RNA , بسبب وجود Complementary stretches ممكن تخليه يكون رابطة هيدروجينية ويعمل شكل ثلاثي الابعاد بنهائيتين 3 و 5 .

من عند منطقة 3 منقدر نعتبرها المكان ارتباط امينو اسيد , على امتداده (الانتشاء يلي تحت) هاد ما يسمى بالانتي كودون

Anti-codon : particular triplet nucleotides that base-pair to specific m RNA codon , it is conventionally written 3-----5 to align properly with codon written 5----3

هو عبارة عن 3 نيكليوتيدات يمكنها انا ترتبط بالكودونات m RNA , وكونه حنعرف انه الترجمة حتصير من خمسة الى ثلاث , إذا حينكتب من ثلاث لخمسة .

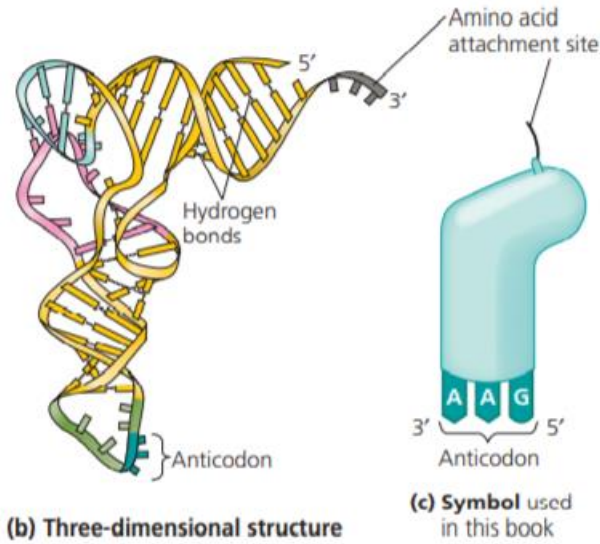
QUESTION : consider m RNA codon 5-GGC -3 , the t RNA that base pair with this codon by hydrogen bonding has ?

- 1) 3-CCG-5 2) 5-CCG-3 3) 3-CGC-5 4) 5-CGC-3

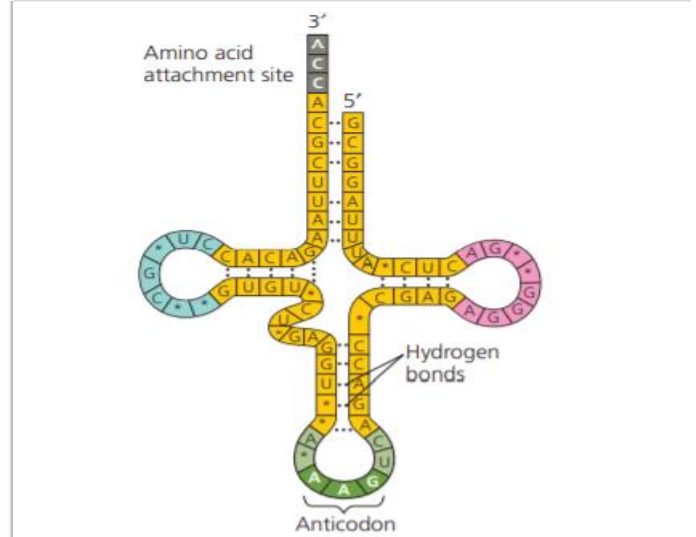
- Like m RNA and other types of cellular RNA , transfer RNA are transcribed from DNA templates . In eukaryotic cell, t RNA like, m RNA , is made in the nucleus and then travels to the cytoplasm . In eukaryotes and bacteria each t RNA molecule is used repeatedly , pick up the designated amino acid in the cytosol and then put it in the growing polypeptides and return to the cytoplasm to pick up another appropriate m RNA .

هون السؤال T RNA من وين طلع كيف تصنع ؟ بما انه من أنواع RNA إذا
من DNA وبحقيقات النواة بتصنع داخل النواة وبعدين بيطلع للسيتوبلازم .
هلا بحقيقات النواة والبكتيريا t RNA منستخدمه بشكل متكرر
بتحمل امينو اسيد يلي بدها ياه وبتضيفه لباقي السلسلة وبترجع مرة ثانية للسيتوسول .

الصورة مهمة



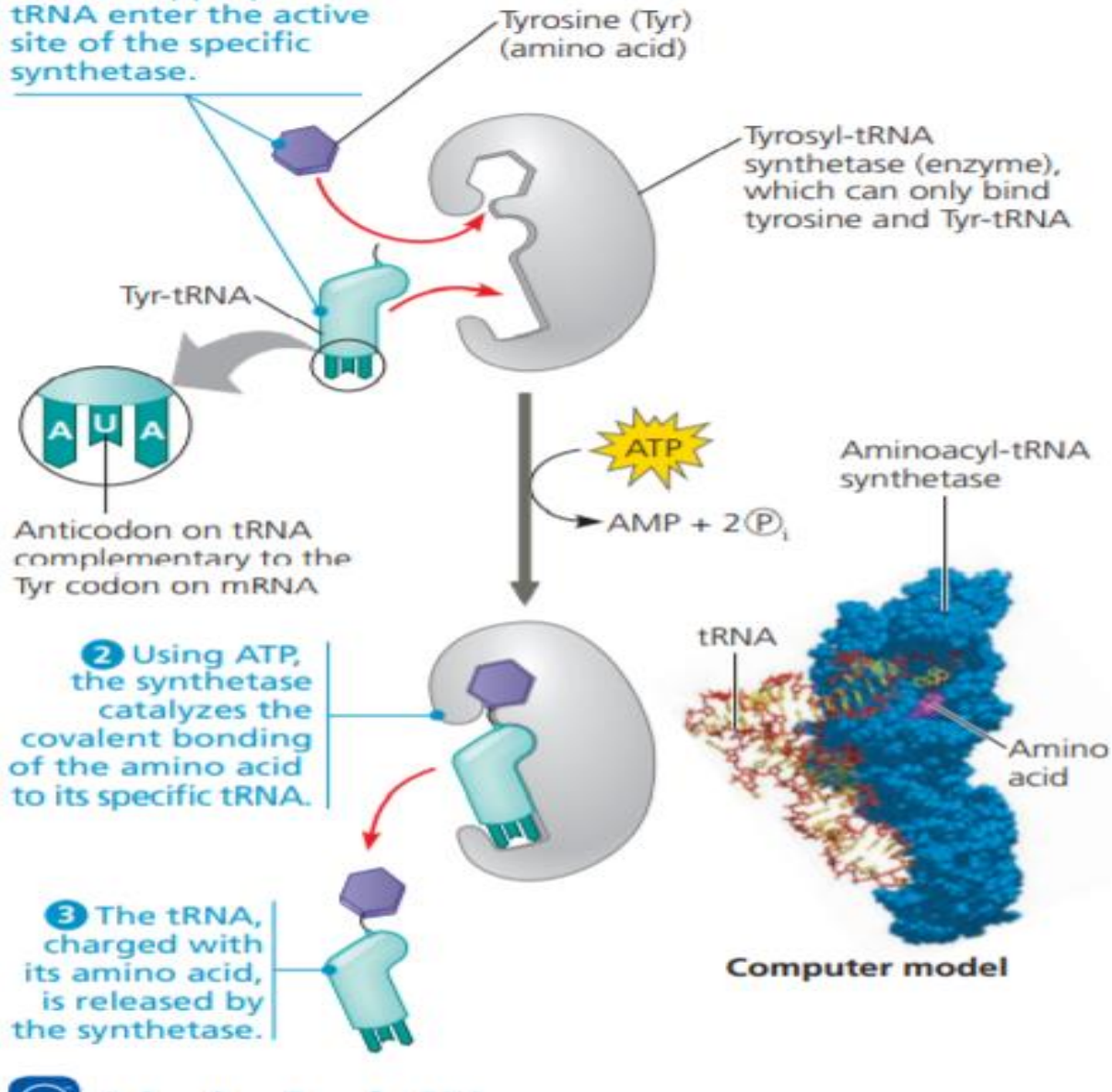
(c) Symbol used in this book



TO ACCURATE TRANSLATION OF A GENETIC MESSAGE REQUIRES TWO INSTANCES OF MOLECULAR RECOGNITION .

▼ **Figure 17.17 Aminoacyl-tRNA synthetases provide specificity in joining amino acids to their tRNAs.** Linkage of a tRNA to its amino acid is an endergonic process that occurs at the expense of ATP, which loses two phosphate groups, becoming AMP.

❶ The amino acid and the appropriate tRNA enter the active site of the specific synthetase.



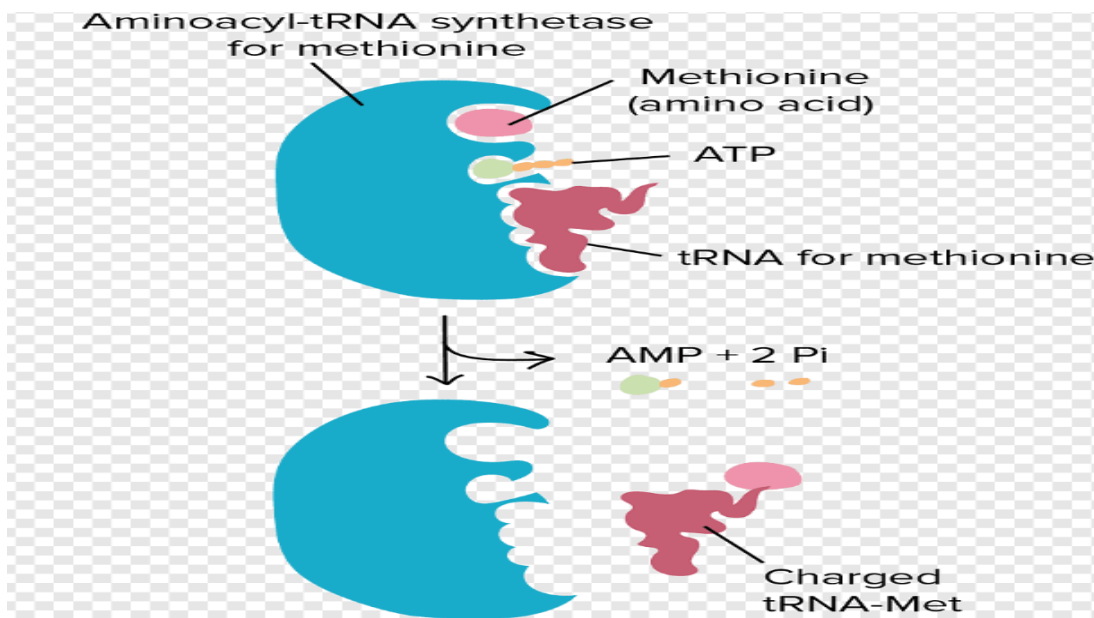
First , t-RNA that binds to an m-RNA codon specifying particular amino acid must carry that amino acid , and no other, to the ribosome . The correct matching up of t-RNA and amino acid is carried out by a family of related enzymes called aminoacyl-t-RNA synthetases . The active site of each type aminoacyl-t RNA synthetase fits only a specific combination of amino acid and t-RNA . There are 20 different synthetase , one for each amino acid . The synthetase catalyze the covalent attachment of the amino acid to its t-RNA in a process driven by the hydrolysis of ATP . The resulting aminoacyl t-RNA , also called charged t-RNA , is released from the enzyme and is available to deliver its amino acid to a growing polypeptide chain on a ribosome .

هلا كل t-RNA يرتبط مع امينو اسيد محدد
aminoacyl-t-RNA synthetases
active site
طيب كيف يرتبطوا ؟ عن طريق مساعدة انزيم

ATP and it called (t RNA and amino acid)
بسرع من تكوين رابطة بينهم بتفاعل يتم في استهلاك

Charged t-RNA and aminoacyl t-RNA

بعدين بطلع
Charged t-RNA from Synthetase
من الانزيم وبصير جاهز انه يروح للرايبوسوم .



أسألکم سؤال ؟؟؟!! هلأ صح انا ویاکم منعرف انه t-RNA حیرتبط m-RNA

كيف ؟ عن طريق

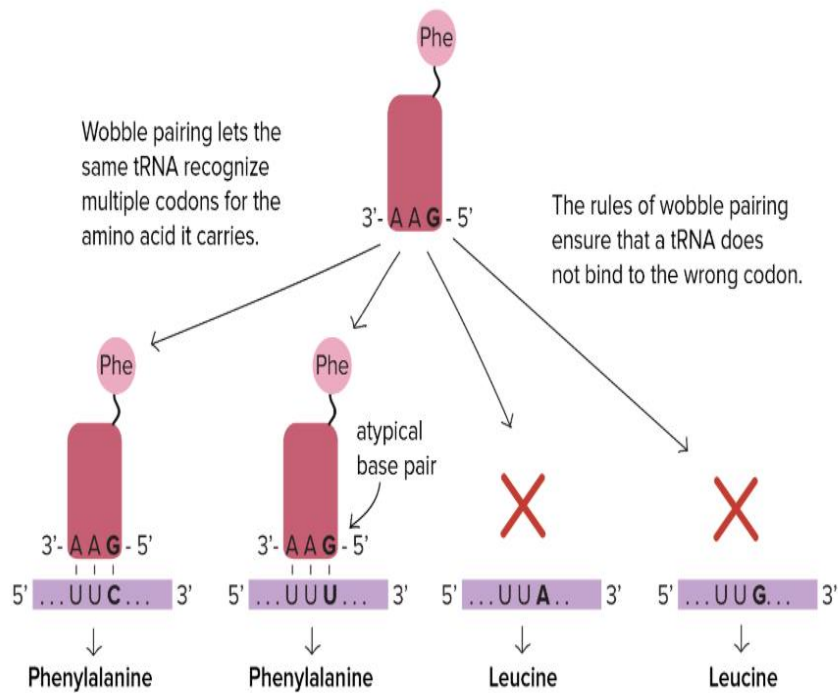
BASE PAIR RULES

اوووعكم تكونوا ناسيين ,

طبيب إذا حكيتكم انه عنا - 45 t RNA وهو مفروض لو بدنا نطبق قواعد

حرفيااااا يطلع 61 t-RNA شو السبب ؟

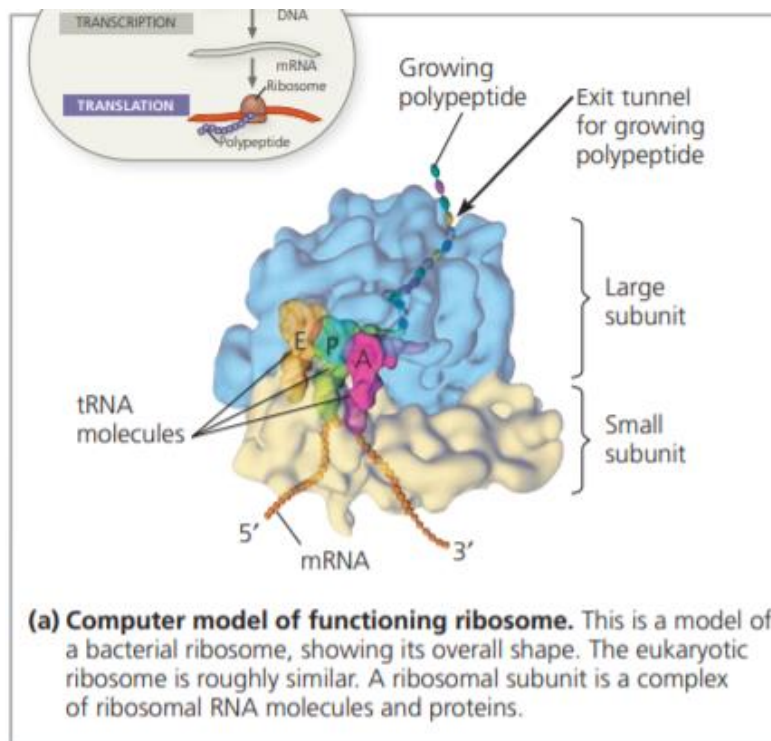
Wobble : Wobble pairing doesn't follow normal rules, but it does have its own rules. For instance, a G in the anticodon can pair with a C or U (but not an A or G) in the third position of the codon .



NOW WE WILL TALK ABOUT THE SITE OF TRANSLATION : RIBOSOMES

RIBOSOMES :consists of a large and small subunits are made from proteins and ribosomal RNA, the large subunit sits on top of the small subunit only when attached to an m-RNA molecule,,and the mass of the proteins is one-third of the ribosome and the rest is rRNAs in bacteria or four in eukaryotes .

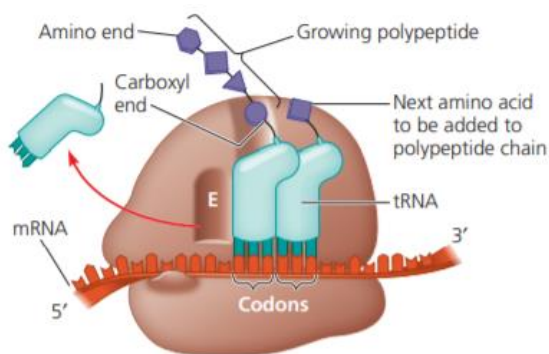
رايبوسومات بتتكون من بروتينات وهي تعتبر المكان التي تحدث فيه الترجمة , هالأ لو اطلعنا عتركيب الرايبوسومات حلاحظ انه عنازي قطعة كبيرة وقطعة صغيرة تمام ؟ هذول ما بيحوا مع بعض الا وقت عملية الترجمة انتبهوا!!!! مش بكونوا هيكل من الأصل .وكتلة البروتينات 3/1 وما تبقي تعتبر كتلة ال r-RNA



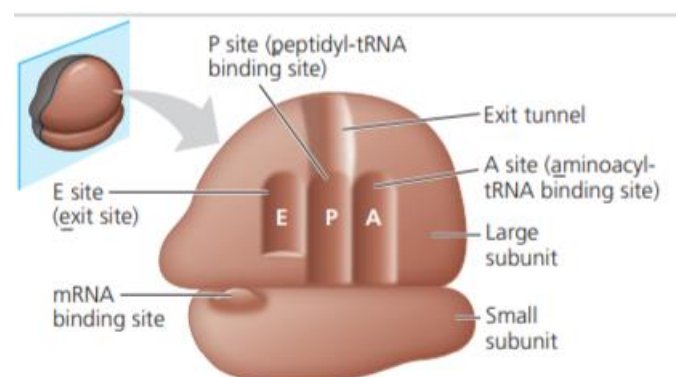
Although the ribosome of bacteria and eukaryotes is very similar in structure and function , eukaryotic ribosomes are slightly larger , as well as differing from bacterial ribosome in their molecular composition . The difference is **medically significant** . Some antibiotic can affect bacterial ribosome not eukaryotic like : tetracycline , strepto-mycine .

صح أنه ريبوسومات البكتيريا وحقيقيات النواة كثير بشبهوا بعض بس حقيقيات النوى أكبر وكمال باختلافها بتركيبها الجزيئي و لهيك هاد الاشياء طبيا كثير مفيد بسبب أنه قدرنا نصنع انتي بيوتيك يآثر ع ريبوسوم البكتيريا مش حقيقيات النواة وهيك البكتيريا ما بتصنع بروتينات وقل تأثيرها

The structure of a ribosome reflects its function of bringing mRNA together with tRNAs carrying amino acids. In addition to a binding site for mRNA, each ribosome has three binding sites for tRNA (**Figure 17.18**). The **P site** (peptidyl-tRNA binding site) holds the tRNA carrying the growing polypeptide chain, while the **A site** (aminoacyl-tRNA binding site) holds the tRNA carrying the next amino acid to be added to the chain. Discharged tRNAs leave the ribosome from the **E site** (exit site).



(c) Schematic model with mRNA and tRNA. A tRNA fits into a binding site when its anticodon base-pairs with an mRNA codon. The P site holds the tRNA attached to the growing polypeptide. The A site holds the tRNA carrying the next amino acid to be added to the polypeptide chain. Discharged tRNAs leave from the E site. The polypeptide grows at its carboxyl end.



(b) Schematic model showing binding sites. A ribosome has an mRNA binding site and three tRNA binding sites, known as the A, P, and E sites. This schematic ribosome will appear in later diagrams.

Ribosomal RNA :is a type of non-coding RNA which is the primary component of ribosomes, essential to all cells.

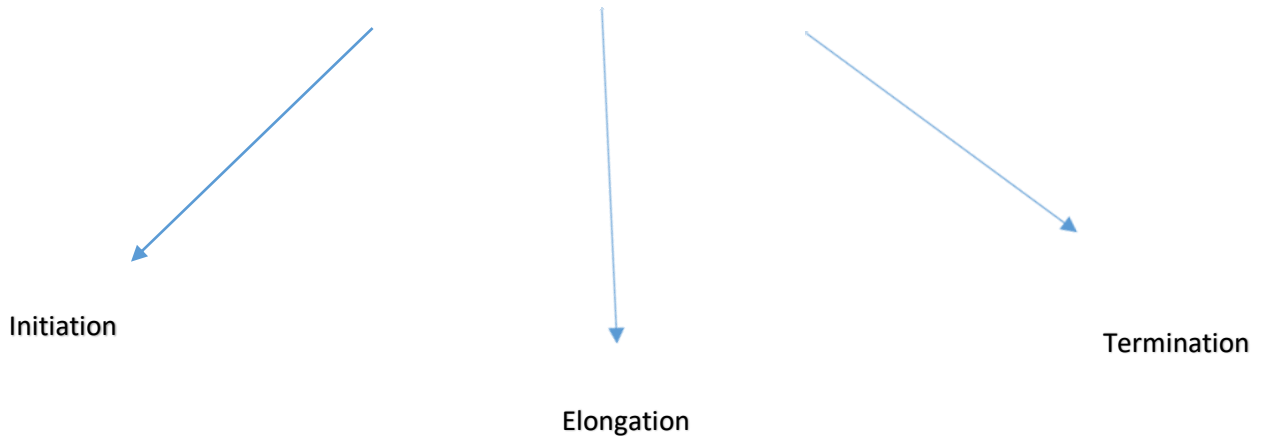
Why the ribosome could actually be considered one colossal ribozyme !

Because they are made from r-RNA and proteins , are primary responsible for both structure and the function of the ribosome . The proteins , which are largely on the exterior , support the shape changes of the rRNA molecules as they carry out catalysis during translation . Ribosomal RNA is main constituent of the A and P sites and of the interface between the two subunits and also as the catalyst peptide formation .

نحن متقدر نعتبر رايبوسومات رايبوزايمات ليه ؟ هلا الأكثر رايبوسومات مصنوع من r-RNA

ومنقدر نحكي بانه r-RNA مكون لمنطقة A and P والمنطقة يلي بين القطعة الكبيرة والصغيرة ويتساعد بتكوين رابطة ببتيدية بينهم , بينما البروتينات بتدعم الشكل والتغيرات يلي حتصير فيه أثناء عملية Translation

BUILDING A POLYPEPTIDE



INITIATION : In either **bacteria or eukaryotes** , the start codon **AUG** , signals to start translation this is important because it establish the codon reading frame for the m-RNA .IN the first step of translation , the small ribosome subunit binds to both m-RNA and specific initiation t-RNA , so that the first components that will associates during the initiation stage of translation are m-RNA , t-RNA – which is bearing the first amino acid (Met) of the polypeptide – and the small ribosomal subunit after that the large subunit will come to complete the **Translation initiation complex** , and the proteins which bring all these component together are called **initiation factors** . In this step GTP will be hydrolyzed to form this complex . At the completion of this step t-RNA will set in P site of the ribosome and the A site will be ready for the next aminoacyl t-RNA .

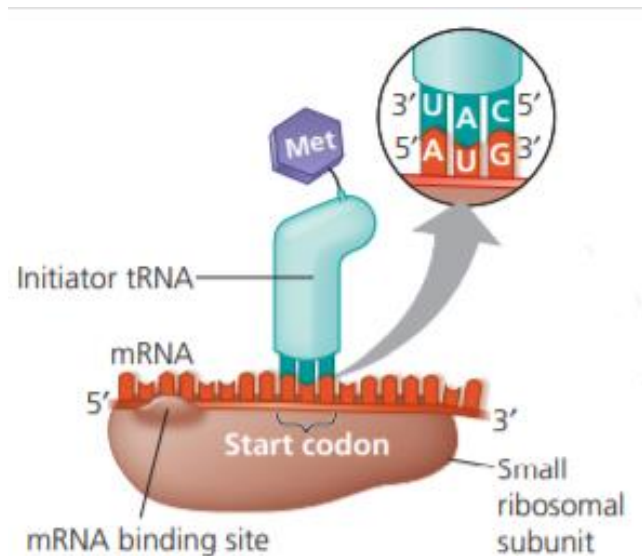
سواء كانت بكتيريا أو حقيقيات النوى ، كودون البدء يعتبر الإشارة لحتى تبدأ عملية الترجمة . هلاً احنا عرفنا انه الريبوسوم ما بتصنع مركب من
 قطعتين بتركب أثناء عملية الترجمة بالأول بتركب القطعة صغيرة و m-RNA and t-RNA

بعدها يتنجي القطعة الكبيرة ويتكون Translation initiation complex ، هاد طبعا بحاجة لبروتينات (ابطال العملية) اسمهم

Initiation factors ليحيوا كل هاء ، المكونات و طبعا حتم استهلاك طاقة ، وبهاي الحالة يكون t-RNA موجود بموقعه الصح
 و P A يستنى aminoacyl t-RNA بيحمل الامينو اسيد الثاني .

There are some differences in binding small ribosome to m-RNA and t-RNA ? SO lets discuss them in details

The **bacterial cell**, the **m-RNA binding site** on the small subunit will recognize a specific nucleotide sequence in the m-RNA just upstream of the start codon . AN the initiator t-RNA sequence , with anti codon UAC , base pair with the start codon ,AUG , This t-RNA carries the first amino acid Met .



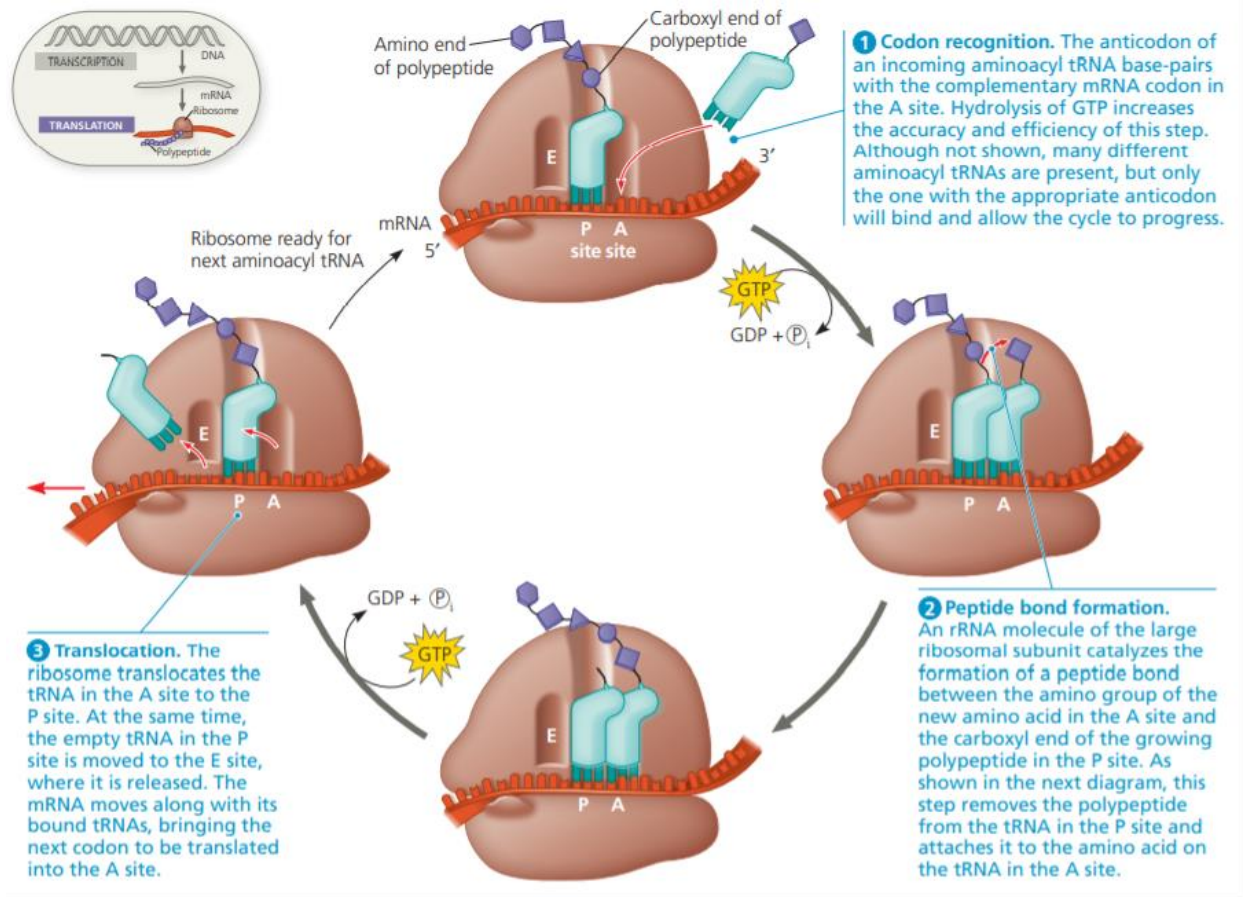
Eukaryotes , the small subunit with the **initiator t-RNA** already bound , binds to the **5'cap** of the m-RNA and then moves or scans , downstream along the m-RNA until it reaches the start codon , the initiator t-RNA then hydrogen bonds to **AUG** start codon .

في الخلية البكتيرية , يرتبط بنيكليوتيدات محددة قبل كودون البدء , بعدين بيحي m-RNA binding site يرتبط بالقطعة الصغيرة وهاد m-RNA binding يرتبط بكودون البدء ويكون حامل الامينو اسيد. بتعرف و

t-RNA and small subunit

اما في حقيقيات النوى , القطعة الصغيرة و t-RNA يرتبطوا بعدها سوا يرتبطوا ب 5'cap بعدها يتحركوا ويعملوا سكان لحتى يوصلوا لكودون البدء , وبعديها t-RNA يكون روابط هيدروجينية مع كودون البدء

ELONGATION : ("middle"): in this stage, amino acids are brought to the ribosome by tRNAs and linked together to form a chain, so its become longer and they add it in C-terminus not N-terminus .



الخطوة الثانية من الترجمة هي زيادة طول سلسلة الامينو أسيد وتتم عن طريق 3 خطوات :

أولا Anticodon Aminoacyl t-RNA رح يرتبط على الكودون في منطقة A in m-RNA وأكيد هاي العملية حتى تكون صحيحة بتحتاج استهلاك طاقة حتى تكون أكثر دقة .

ثانيا بدنا نكون رابطة ببتيدية ف r-RNA يلي موجود بالقطعة الكبيرة حيسرع هاي العملية , هلا انتبهوا للصورة السلسلة زي عم تنقطع السلسلة الطويلة يلي بالمنطقة P وترتبط بالامينو اسيد يلي c-terminus in A , وفي طبعا كمان بهاي الخطوة استهلاك جزيئين طاقة الجزيء الأول للدقة والثاني حتى زي يتزود بالطاقة للخطوة الثالثة .

ثالثا هلا الريبوسوم يحرك t-RNA من A إلى P , وبنفس الوقت ال t-RNA يلي ما بيحتوي عالسلسلة حيتحرك من p ل E حتى يخرج ويرجع للسيتوسول و بهيك بترجع منطقة A فاضية مستعدة تستقبل أي t-RNA بيحمل امينو اسيد جديد .

In this step , we use two GTP one for making this in efficient way and the another for store it to another step . (we use GTP not ATP because GTP is more higher energy than ATP).

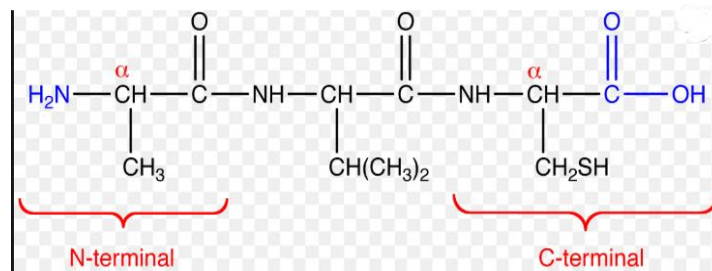
The proteins that help us to add the amino acid in to the chain to be longer are called **Elongation factors**.

REMEMBER THE C AND N -TERMINUS

The C-terminus (also known as the carboxyl-terminus,) is the end of an amino acid chain (protein or polypeptide), terminated by a free carboxyl group (-COOH).

The N-terminus (also known as the amino-terminus,)is the start of a protein or polypeptide referring to the free amine group (-NH₂) located at the end of a polypeptide.

When the protein is translated from messenger RNA, it is created from N-terminus to C-terminus. The convention for writing peptide sequences is to put the C-terminal end on the right and **write the sequence from N- to C-terminus**.



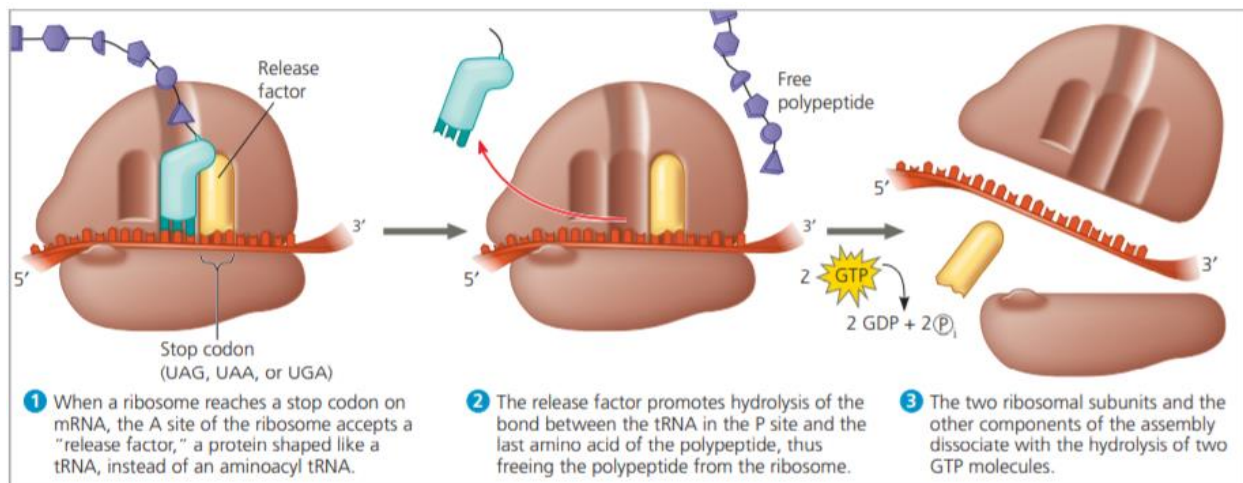
TERMINATION : in the last stage, the finished polypeptide is released to go and do its job in the cell. Termination happens when a stop codon in the mRNA (UAA, UAG, or UGA) enters the A site.

Stop codons are recognized by proteins called release factors, which fit to the P site (though they aren't tRNAs). They add a water molecule to the last amino acid of the chain. This reaction separates the chain from the tRNA, and the polypeptide is released from the **exit tunnel** .

Breakdown of translation assembly requires the hydrolysis of two or more GTP .

آخر خطوة بعملية الترجمة التي يتم فيها اخراج سلسلة الببتيد الى الخلية للقيام بواجبها , عندما يصل الريبوسوم الى كودون الوقف بالتا Release factor بيحي عند A بدال Aminoacyl -tRNA ولهيك هاد الاشئ بحفز قطع الرابطة الببتيدية في منطقة p عن طريق إضافة الماء في اخر امينو اسيد .

ولهيك القطعتين والمكونات الثانية بنفكوا ويتم استهلاك جزيئين طاقة لاتمام هذه العملية .



COMPLETING AND TARGETING THE FUNCTIONAL PROTEIN.

During its synthesis, a polypeptide chain begins to coil and fold spontaneously as a consequence of its amino acid sequence (primary structure), forming a protein with a specific shape: a three-dimensional molecule with secondary and tertiary structure. Thus, a gene determines primary structure, which in turn determines shape.

Additional steps—*post-translational modifications*—may be required before the protein can begin doing its particular job in the cell. Certain amino acids may be chemically modified by the attachment of sugars, lipids, phosphate groups, or other additions. Enzymes may remove one or more amino acids from the leading (amino) end of the polypeptide chain. In some cases, a polypeptide chain may be enzymatically cleaved into two or more pieces. In other cases, two or more polypeptides that are synthesized separately may come together, if the protein has quaternary structure; an example is hemoglobin.

بالإضافة إلى ذلك ، فإننا نلاحظ أن سلسلة من الأمينات (تذكرنا شابتير 5 بيو 101) هالاي شو مش بروتين لسا ما بتعمل وظائف ليهك هلا لازم تنطوي بشكل ثلاثي الأبعاد حتى تعمل بروتين بالإضافة لتعديلات كيميائية على الأمينات أسيد مثل إضافة الدهون أو الكربوهيدرات والمجموعة الفوسفاتية ، ممكن يتم إزالة أمين أسيد واحد أو أكثر من واحد عن طريق الانزيمات . وفي بعض الحالات يتم قطع السلسلة من الأمينات أسيد إلى قطعتين أو أكثر وفي حالات أخرى يتم دمج أكثر من سلسلة معا .

Targeting polypeptide to Specific location

Remember:

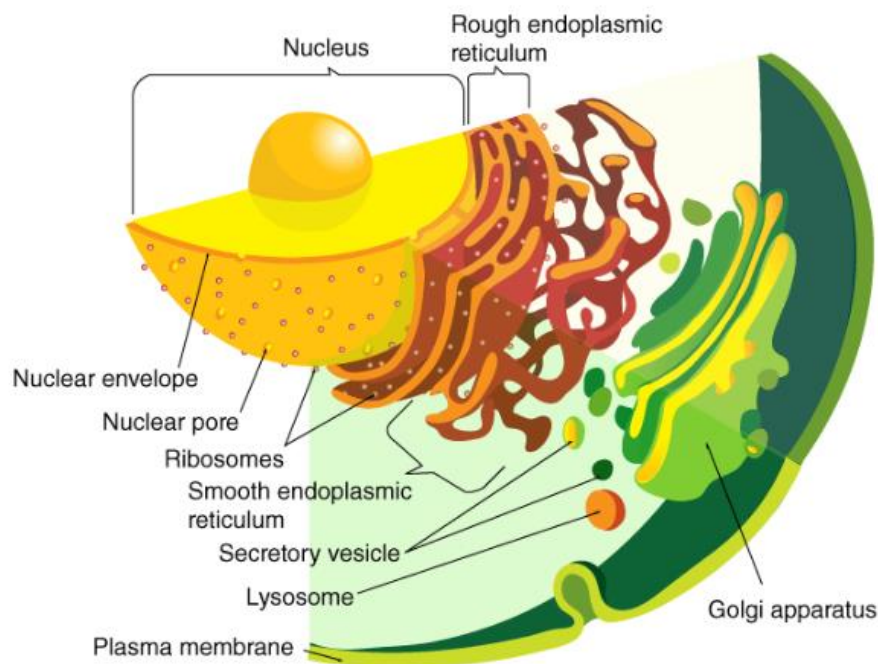
There are **two population** of ribosomes : free and bound ribosomes , that ribosomes themselves are identical but they could alternates between free and bound ribosomes .

Free ribosome : are type of ribosome which are suspended in the cytosol and usually synthesize proteins that stay in the cytosol and function there .

Bound ribosome : type of ribosome which makes proteins for the endomembrane system as well as proteins secreted from the cell.

endomembrane system is : endoplasmic reticulum, Golgi apparatus, vesicles and cell membrane and nuclear envelope.

ENDOMEMBRANE SYSTEM

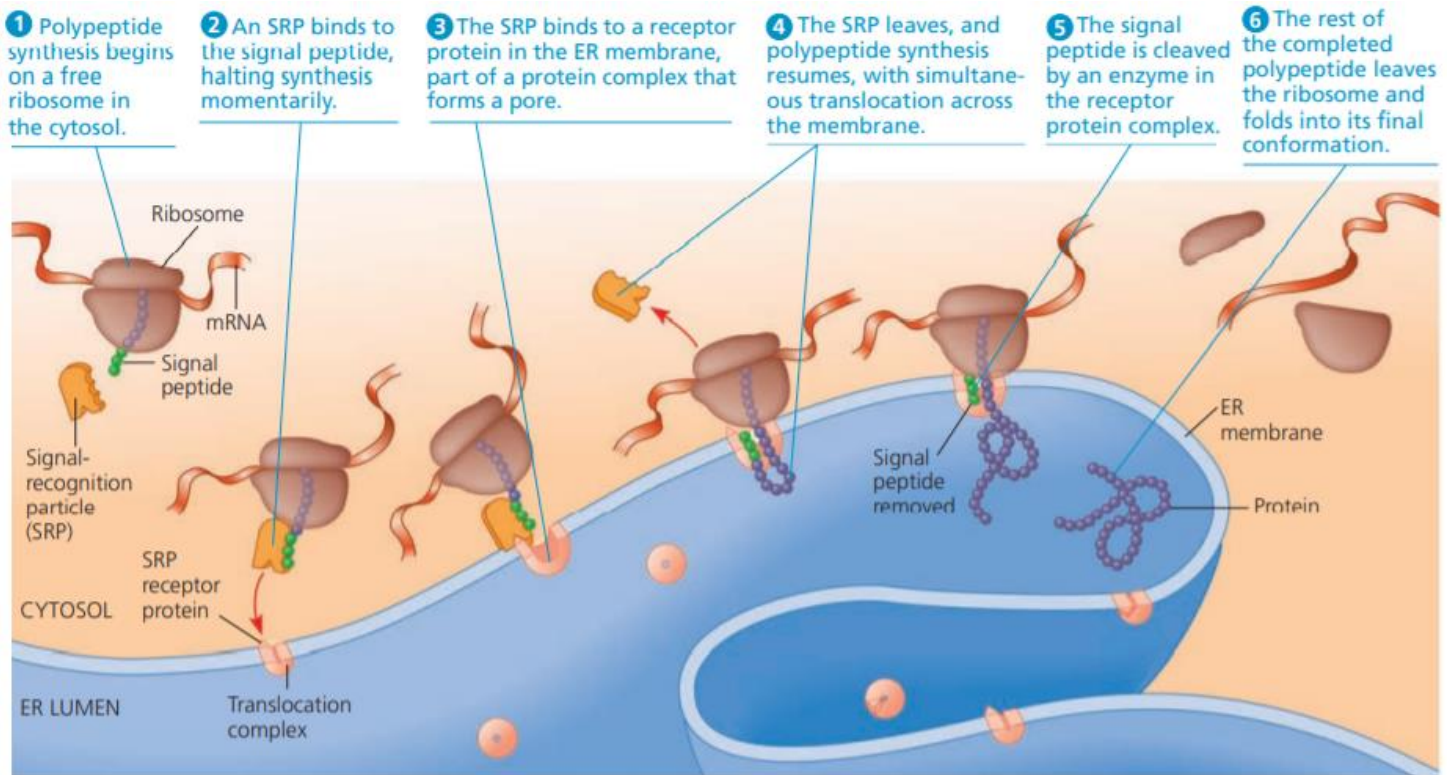


Polypeptides synthesis always begins in the cytosol as free ribosome starts to translate an m-RNA molecule . There , the process continues to completion unless the growing polypeptide itself cues the ribosome to attach to the ER. The polypeptides of proteins destined for the endomembrane system or secretion are marked by a **signal peptide** , which target the protein to the ER. Signal peptide , sequence of about 20 amino acid or near the leading end N-terminus of the polypeptides is recognized as it emerges from the ribosome by protein –RNA complex is called **signal recognition particle** (SRP) . This particle escorts the ribosome to a receptor protein built built into the ER membrane . The receptor is part of multiprotein translocation complex . Polypeptide synthesis continues there , and the growing polypeptide snakes across the membrane into the ER lumen so the polypeptide could remain in the ER or travels by vesicles to other destination .

هون منحكي انه سلسلة الببتيد دائما صنعها يبدأ في السيتسول من ريوسومات الحرة يلي ما بتكون مرتبطة بالشبكة الاندوبلازمية لعاااااااا شو يلي بخلي الريوسوم يرتبط بالشبكة الاندبلازمية وتصير سلسلة الأميно اسيد موجودة في تجويف الشبكة الاندوبلازمية ؟ نوضح أكثر هاد النوع من سلسلة الببتيد بيحتوي باخره زي إشارة signal من خلاله منعرف انه هاى السلسلة الامينو اسيد حتروح للشبكة الاندوبلازمية.

تتكون من تقريبا من 20 امينو اسيد يكون واقع في نهاية الريبوسوم ويتم التعرف عليها من خلال مركب مصنوع من بروتينات و RNA اسمه Signal recognition . هاد بس يرتبط في حيوديه لمستقبل موجود على غلاف الشبكة الاندوبلازمية الملساء ويرتبط في partical ويكمل تكون السلسلة وبس يخلص صنع سلسلة الامينو اسيد هووون السلسلة يا بتضل موجودة بتجوف الشبكة الاندوبلازمية وبتصير أحد بروتيناتها أو بتسافر لاماكن ثانية .

بـعـرف أنـكـم انـعـجـقـتـمـا بـس كـان لـازـم اـحـط فـقـرة الـكـتـاب هـلا نـركـز انا وياكـم بـالـصـورة والـخـطـوات يـلي تـحت
تـمـااااا



أولا صناعة سلسلة الامينو أسيد تبدأ من رايبوسوم حر في السيتيسول .

ثانيا **srp** حيرتبط **Signal** يؤدي إلى توقف صنع السلسلة **بشكل مؤقت** .

ثالثا **Srp** سترتبط بمستقبل بروتيني موجود على غشاء الشبكة الاندوبلازمية الملساء .

رابعا **srp** حتنقارم وحترجع عملية الترجمة تستمر وحناحظ انه سلسلة الامينو اسيد عم تدخل شوي شوي داخل التجويف

خامسا **Signal** سوف يتم ازلتها من خلال انزيم موجود داخل مستقبل البروتيني

سادسا الريبوسوم خلص بنفك وبتنتهي عملية الترجمة وهون بتصير سلسلة الامينو اسيد بدها تتحول لبروتينات وتقرر إذا بدها تضل جوا الشبكة الاندوبلازمية ولا بدها تروح لاماكن ثانية عن طريق الحويصلات.

MAKING MULTIPLE POLYPEPTIDES IN BACTERIA AND EUKARYOTES

In previous sections, you learned how a single polypeptide is synthesized using the information encoded in an mRNA molecule. When a polypeptide is required in a cell, though, the need is for many copies, not just one.

A single ribosome can make an average-sized polypeptide in less than a minute. In both bacteria and eukaryotes, however, multiple ribosomes translate an mRNA at the same time ; that is, a single mRNA is used to make many copies of a polypeptide simultaneously. Once a ribosome is far enough past the start codon, a second ribosome can attach to the mRNA, eventually resulting in a number of ribosomes trailing along the mRNA. Such strings of ribosomes, called **polyribosomes** (or **polysomes**), can be seen with an electron microscope; they can be either free or bound. They enable a cell to rapidly make many copies of a polypeptide.

Another way both bacteria and eukaryotes augment the number of copies of a polypeptide is by transcribing multiple mRNAs from the same gene.

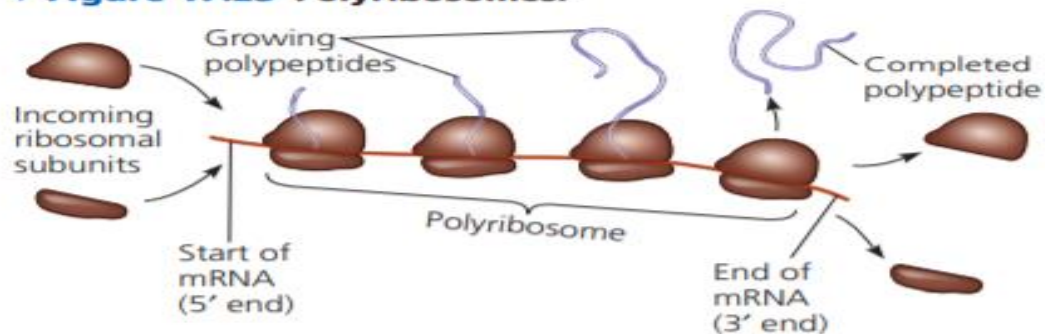
هلاً ريبوسوم واحد بعمل سلسلة امينو اسيد بأقل من دقيقة لكن احنا بدنا نسرع العملية فكيف ؟ ممكن عدة ريبوسومات تترجم

واحد وبالتالي حيطلع عنا نسخ متعددة من سلسلة الامينو اسيد وهاد الاشئ بصير انه هلاً ريبوسوم الأول اربط بكودون البدء وبلش يترجم بس يمشي ويروح للكودون الثاني بعد البدء بيحي ريبوسوم ثاني يرتبط بكودون البدء وهكذا .

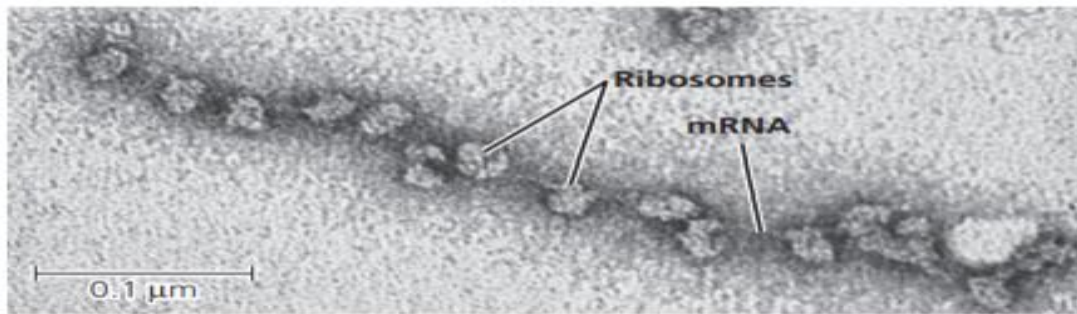
طبعا في طريقة ثانية انه ننتج نسخ كثيرة m-RNA من نفس الجين وهيك بتتصنع كمية أكبر عنا .

POLYRIBOSOME : a group of ribosomes bound to an mRNA molecule like “beads” on a “thread”. [1] It consists of a complex of an mRNA molecule and two or more ribosomes that act to translate mRNA instructions into polypeptides.

▼ **Figure 17.23 Polyribosomes.**



(a) An mRNA molecule is generally translated simultaneously by several ribosomes in clusters called polyribosomes.

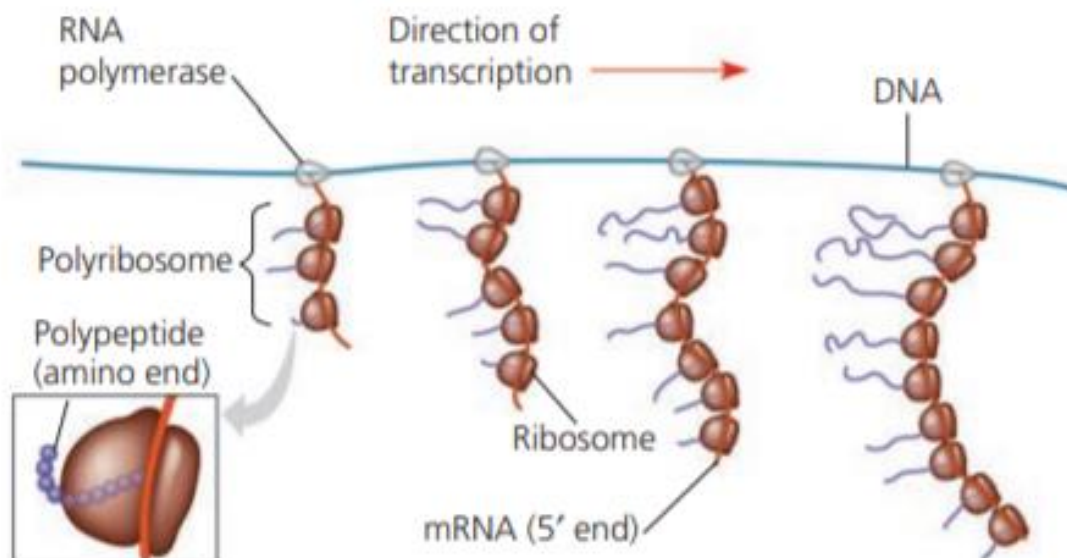
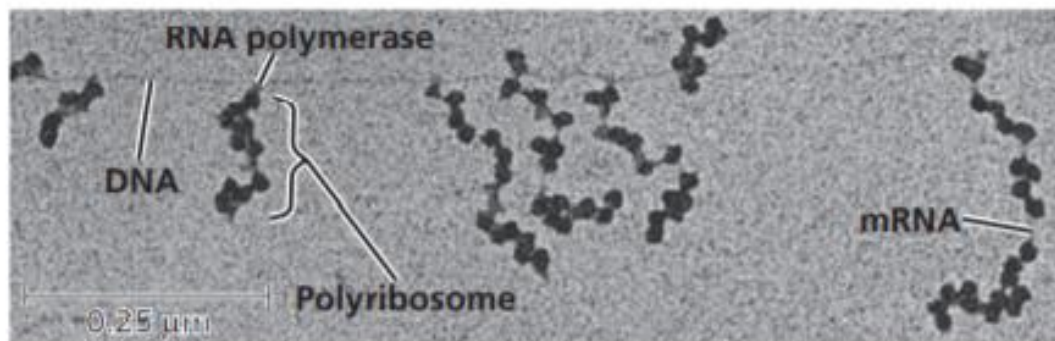


(b) This micrograph shows a large polyribosome in a bacterial cell. Growing polypeptides are not visible here (TEM).

DIFFERENCE IN TRANSLATION AND TRANSCRIPTION BETWEEN BACTERIA AND PROKARYOTES

Prokaryotic transcription occurs in the cell cytoplasm.	Eukaryotic transcription occurs in the cell nucleus.
Here, transcription and translation happen simultaneously.	Transcription and translation differ in space and time.
mRNA is transcribed directly from template DNA molecule.	Initially a pre-mRNA molecule is formed and then processed to yield a mature mRNA.

▼ **Figure 17.24 Coupled transcription and translation in bacteria.** In bacterial cells, the translation of mRNA can begin as soon as the leading (5') end of the mRNA molecule peels away from the DNA template. The micrograph (TEM) shows a strand of *E. coli* DNA being transcribed by RNA polymerase molecules. Attached to each RNA polymerase molecule is a growing strand of mRNA, which is already being translated by ribosomes. The newly synthesized polypeptides are not visible in the micrograph but are shown in the diagram.



a) Bacterium

b) Eukaryote

